
Detecting AI Training in Datacentre Power

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Abstract

Compute-governance regimes that restrict large training runs presuppose that a regulator can tell when training happens on monitored hardware, and a power meter outside the operator’s control is the one passive channel available against an operator who does not cooperate. We ask what such a meter can certify, and answer with a ladder of four claims, each with a stated information cost and a hard ceiling. Evidence comes from a literature-parameterized scenario model of datacentre AI workloads and an explicit off-chip observation model, against both a benign operator and a concealing one who knows the detector. The iteration cadence of synchronous training is recoverable by a tracked cyclostationary detector, but the complete pipeline admits no analytic calibration, so we report it as an effective score with per-trace surrogate calibration rather than a constant false-alarm-rate test. Our central result is the de-periodicisation frontier, which prices hiding against measured throughput anchors and is strongly asymmetric: varying real work per iteration hides from every fixed-frequency test at essentially zero cost, while the tracking class holds until that variation is extreme, where it still detects about half of runs. That extreme level is free in throughput but not in learning: in small-scale training it costs the operator 267% more compute to reach the same loss, and a better-designed variant that hides about three-quarters of runs still costs about 80%, even with a tuned recipe. Two limits are firmer than the detectors. The observation channel is itself a condition on every claim — a one-second-integrating, one-hertz meter defeats all detectors on an honest workload — and the meter certifies a physical schedule, not training semantics: five non-training loads, including one computing gradients and discarding them, are

classified as training by every rule we test. Escaping that ceiling means leaving the passive channel for an active challenge protocol, which we specify but do not run.

1. Introduction

AI governance agreements will require regulating training runs (Shavit, 2023; Sastry et al., 2024; Baker et al., 2025; Heim & Koessler, 2024). This will require an external verifier to identify when training is occurring on a monitored cluster, against an operator who is trying to hide their training. There are multiple channels that could provide the verifier with diagnostic information to identify hidden training. These include on-chip mechanisms that log activities for later analysis, network taps that monitor data exchanged between chips, or human-centred approaches such as whistleblower programs or interviews of operator personnel. However each of these channels requires trusting either hardware that is under the control of the operator, or the operator personnel themselves. Conversely, off-chip sensors that take analog, physical measurements of the cluster (e.g. power or electromagnetic emissions) do not require trusting the operator. Such sensors could be retrofitted to existing datacentres (e.g. clamping a power meter on an input cable) and secured against tampering by the operator.

AI training has already been shown to leave a signature in the power drawn by the hardware that can serve as a diagnostic to identify and separate training from inference. A 10 MHz current probe clamped around the GPU’s PCIe power supply cables can separate training from inference and from non-AI work at 97% accuracy (Gargiulo & Kulp, 2026), a result robust against many but not all adversarial strategies. On-chip 1 Hz telemetry counters that measure power (among other telemetry: utilisation, memory, etc.) can classify training at 95% accuracy, again a result robust against many but not all adversarial strategies (Rahman & Tajdari, 2026). In practice, any realistic AI governance agreements that use off-chip sensors will rarely have such fine-grained access to the GPU hardware. Power meters that can be retrofitted to existing facilities and secured against tampering are likely to sit at the level of the rack or facility input, with sampling frequencies ~ 10 Hz (i.e. three to four orders of magnitude slower than the board probe) and behind

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a power-delivery path that low-passes and dilutes the signal. At this level training can be identified and it is large enough that datacentre and grid operators treat it as an engineering control problem to be dealt with (Patel et al., 2023; Ko & Zhu, 2025).

In this paper we develop a new method for detecting training signatures given noisy facility level power measurements, against an adversarial operator who is trying to hide their training.

We do:

- * specify an off-chip observation model and a literature-parameterised generator of datacentre training and inference power,
- * construct a detector for the cadence that tracks the wandering line, borrowing the technique from signal analysis and GW astrophysics
- * We construct the Pareto frontier and show the trade-off between how much of the cadence a concealing operator removes and what that removal costs in measured throughput (Section 6).
- * We derive a minimum meter specification, the sampling rate, integration time and in-band transfer i.e. the specs needed for a meter to work
- * We restrict attention to synchronous, iteration-structured training observed through a time-resolved trace; the integrated-energy channel, asynchronous schedules and the calibration of the generator to distributed hardware are outside scope, and Section 7.3 says what that costs.

Unlike a covert transmitter, a training operator cannot freely reshape its signal: the communication phase has a fixed physical duration, and every distortion of the cadence costs throughput. It is this constraint that makes a cost-of-hiding bound possible.

This paper is organised as follows. Section 2 gives the model of datacentre training and inference power that generates every trace below, and Section 3 fixes what the verifier sees through an off-chip meter and what a concealing operator may do to hide from it. Section 4 develops the detector, a Viterbi line tracker; Section 5 reports its power against the inference null and the minimum meter it needs; and Section 6 prices what it costs a concealing operator to defeat it. Section 7 asks how likely a concealing operator is to be detected over a run, states what a detection does and does not certify, and sets out the limitations, and Section 8 concludes.

2. A model of datacentre training power

In this section we build a model of the power draw from a data centre running AI workloads. This model will serve as a synthetic data generator to test our methods, in lieu of having access to real empirical data. We adopt the stochastic model of AI training power from Ko & Zhu (2025), developed for the analysis of grid stability, modified to add an Ornstein-Uhlenbeck drift on the training frequency. The reader is referred to Ko & Zhu (2025) for the full training model and its context.

This section is organised as follows. Section 2.1 describes the power draw from a training workload, following Ko & Zhu (2025), and adds our drift of its cadence. Section 2.2 describes the power draw from a fine-tuning workload. Section 2.3 describes the power draw from an inference workload. Section 2.4 combines these workloads into the load of a whole datacentre: one dominant workload, either training or inference, over a background of smaller trainings and fine-tunings. Table 1 collects every parameter value of the model.

2.1. Training

Multi GPU AI training of the kind that typifies frontier models is composed of a two-phase iterative cycle. First there is a power intensive, high compute “up” phase where the forward/backward pass happens. Second there is a low power, low compute, communication “down” phase where the gradient all reduce happens.

Iteration i has period

$$T_i = \frac{1}{f_0 (1 + \xi_i)}, \quad \xi_i \sim \mathcal{N}(0, \sigma_\xi^2), \quad (1)$$

where f_0 is the baseline fluctuation frequency. For a given training run f_0 is fixed. Between training runs f_0 can vary depending on model size, batch size, parallelism layout, interconnect and hardware (Patel et al., 2024a; Jain et al., 2024). To describe the inter-workload variation, we take $f_0 \sim \mathcal{U}(f^{\text{lo}}, f^{\text{hi}})$. The period splits into an up-phase of duration $\rho_i T_i$ and a down-phase of duration $(1 - \rho_i) T_i$, with the phase ratio $\rho_i \sim \mathcal{U}(\rho^{\text{lo}}, \rho^{\text{hi}})$ redrawn each iteration. Writing t for time within iteration i , $P(t)$ for the power the run draws above the facility’s static overhead (added in Section 3.1), and P_{peak} for its peak,

$$\frac{P(t)}{P_{\text{peak}}} = \begin{cases} 1 + \Delta_i^{\text{up}} + \nu^{\text{up}}(t), & t < \rho_i T_i, \\ 1 - \Delta_i^{\text{dn}} + \nu^{\text{dn}}(t), & t \geq \rho_i T_i, \end{cases} \quad (2)$$

where the per-phase level offsets are $\Delta_i^{\text{up}} \sim \mathcal{N}(0, \sigma_\Delta^2)$ and $\Delta_i^{\text{dn}} \sim \mathcal{N}(\mu_\Delta, \sigma_\Delta^2)$, and $\nu^{\text{up}}, \nu^{\text{dn}}$ are zero-mean intra-phase Gaussian fluctuations with standard deviations $\kappa_\nu s_i^{\text{up}}$ and $\kappa_\nu s_i^{\text{dn}}$. The s_i^{up} and s_i^{dn} are redrawn each phase, uniformly from the ranges $[s^{\text{up,lo}}, s^{\text{up,hi}}]$ and $[s^{\text{dn,lo}}, s^{\text{dn,hi}}]$,

and κ_ν is a fixed scale factor. Table 1 lists the values of every parameter in this section.

Equations (1) and (2) are the exact training workload model from Ko & Zhu (2025). We make one modification, adding a slow drift to the cadence f_0 itself. The jitter ξ_i does not produce drift. Because it is redrawn independently each iteration, the cadence averaged over many iterations is still f_0 . The timing errors accumulate as a random walk in phase, which broadens the spectral line into a Lorentzian of full width $\approx 2\pi\sigma_\xi^2 f_0$ (to leading order in σ_ξ) but leaves its centre at f_0 . Real runs vary differently. Specifically, the iteration time changes in ways that persist. Every training step waits for the slowest GPU, so one slow component slows the whole run. In a production LLM cluster, these slowdowns last for the whole job rather than varying from step to step (Lin et al., 2025). Hardware can also keep running but degrade: CPU contention, overheating GPUs and congested network links cut throughput by tens of percent for periods of tens of seconds to hours (Wu et al., 2024). Over a day, temperature shifts throughput by a further 1–2% (Grattafiori et al., 2024).

We model this by letting the cadence of iteration i be $f_0 g_i$, where g_i is a discrete Ornstein–Uhlenbeck (first-order autoregressive) process that starts at $g_0 = 1$ and reverts to 1:

$$g_i - 1 = (1 - \theta)(g_{i-1} - 1) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_g^2). \quad (3)$$

The period of Equation (1) becomes

$$T_i = \frac{1}{f_0 g_i (1 + \xi_i)}, \quad (4)$$

which reduces to Equation (1) when $g_i \equiv 1$. The stationary variance of Equation (3) is $\sigma_g^2 / (2\theta - \theta^2)$, so we set

$$\sigma_g = \frac{\sigma_f}{f_0} \sqrt{2\theta - \theta^2}, \quad (5)$$

which makes the stationary standard deviation of the instantaneous cadence $f_0 g_i$ equal to σ_f , in Hz. The autocorrelation of g_i at lag k is $(1 - \theta)^k$, so the cadence stays correlated over $\sim 1/\theta$ iterations. Over that span it wanders by σ_f , which can exceed both the line width set by ξ_i and the frequency resolution of a trace. We clip g_i from below at $g_{\min}$ so the period stays finite. We hold θ fixed and treat σ_f as a free parameter, sweeping it in Section 5.

2.2. Fine-tuning

Fine-tuning workloads can be represented using the same structure as in Section 2.1, with different parameters (Table 1). We denote the fine-tuning frequency as f_1 . Compared with training, it has a lower cadence band, a longer tail phase (larger ρ_i), a deeper tail-to-idle swing (larger μ_Δ),

and slightly smaller level offsets and intra-phase noise. We apply the cadence drift of Equation (3) only to the dominant run; fine-tuning and background training runs keep a fixed cadence ($\sigma_f = 0$).

2.3. Inference

Ko & Zhu (2025) do not present a model for power draw of inference workloads. We now construct one. We suppose three structures on a constant mean serving power load B . First, token generation (decode). Serving systems use continuous batching, which adds and removes requests from the batch at every step (Yu et al., 2022; Kwon et al., 2023). A per-step cadence exists, but the batch composition reshuffles every step. Unlike training, there is therefore no characteristic frequency: decode power smears across a broad band with no coherent frequency path. We model it as band-limited noise $b(t)$ of unit RMS over $[f_b^{\text{lo}}, f_b^{\text{hi}}]$, scaled by an amplitude A . Second, expert-load imbalance. Under sparse mixture-of-experts routing, tokens are not spread evenly across the experts (Shazeer et al., 2017), so the work per step varies with the mix of requests. We model this as a slow envelope on the decode power: a unit-variance Ornstein–Uhlenbeck process $u(t)$ with correlation time τ_u , entering at depth m . Amplitude modulation spreads the decode energy into sidebands, which makes the null harder, not easier. Third, prompt processing (prefill). Each request begins with a prompt phase that is brief, compute-bound, and draws more power than token generation (Patel et al., 2024a;b). We model these phases as box pulses $s(\cdot)$ of duration d_p and height a , arriving at times τ_k from a Poisson process of rate λ_p . These memoryless arrivals carry no line and simply raise the broadband in-band level. Together,

$$\frac{P^{\text{inf}}(t)}{P_{\text{peak}}} = B + A [1 + m u(t)] b(t) + \sum_k a s(t - \tau_k). \quad (6)$$

The forms of these structures follow how LLM serving systems schedule work; their amplitudes are our choice (Table 1). We set them so that inference is a loud confound rather than a quiet one. As Figure 1 shows, it delivers more in-band power than training but an order of magnitude less spectral contrast.

2.4. Aggregate

A datacentre rarely runs one job at a time. Following Ko & Zhu (2025) we write the load behind a single meter as the aggregate of one dominant training run over a background of smaller trainings and fine-tunings,

$$P_{\text{AI}}(t) = P^{\text{main}}(t) + \sum_{j=1}^{N_{\text{tr}}} P_j^{\text{tr}}(t) + \sum_{k=1}^{N_{\text{ft}}} P_k^{\text{ft}}(t). \quad (7)$$

where P^{main} is the power of the dominant training run, P_j^{tr} that of the j -th of N_{tr} smaller training runs, and P_k^{ft} that of the k -th of N_{ft} fine-tuning runs. Each term is an independent draw of Equation (2), with its own cadence, phase ratios, and level offsets. P^{main} and the P_j^{tr} use the training parameters and the P_k^{ft} use the fine-tuning parameters; only P^{main} carries the cadence drift of Equation (3). Peak power is split between the dominant run, the small trainings, and the fine-tunings in the ratio $w_{\text{main}} : w_{\text{tr}} : w_{\text{ft}}$, and each background share is divided equally among its members. Each small training draws its cadence from the same band as the dominant run, so the background carries weak lines of its own inside the search band.

We define the null aggregate as the same datacentre with the dominant training run replaced by inference (Section 2.3). The background is unchanged,

$$P_{\text{AI}}^{\text{null}}(t) = P^{\text{inf}}(t) + \sum_{j=1}^{N_{\text{tr}}} P_j^{\text{tr}}(t) + \sum_{k=1}^{N_{\text{ft}}} P_k^{\text{ft}}(t), \quad (8)$$

with P^{inf} from Equation (6) at the same share w_{main} . The two aggregate classes are then power-matched and both contain the background’s weak lines. Figure 1 shows the four resulting workload classes as a verifier would see them.

3. Measurement and threat model

This section specifies how the power drawn by the data centre reaches the verifier (i.e. the measurement model), and what the facility operator can control (i.e. the threat model). The verifier wants to know whether a monitored facility is running undeclared training. We frame this as a hypothesis test. Under the null hypothesis $\mathcal{H}_0$ the facility runs its declared workload, which for a monitored cluster is dominated by inference: the inference model of Section 2.3, alone or over the background of Equation (8). Under the alternative hypothesis $\mathcal{H}_1$ a training run of Section 2.1 takes the place of that inference, alone or over the same background (Equation (7)). Section 3.1 outlines the channel through which either hypothesis reaches the verifier. Section 3.2 outlines who controls the load, what they know, and what they may change.

3.1. The measurement model

Section 2 gives the power $P_{\text{AI}}(t)$ drawn by the AI workload above the facility’s static overhead, in units of P_{peak} . The total power behind the meter is

$$P_{\text{fac}}(t) = P_0 + P_{\text{AI}}(t), \quad (9)$$

with P_{AI} given by Equation (7) under $\mathcal{H}_1$, by Equation (8) under $\mathcal{H}_0$, or by a single workload of Section 2 when a result drops the background. The verifier holds one scalar

	Symbol	Train	Fine-tune
Cadence band [Hz]	$[f^{\text{lo}}, f^{\text{hi}}]$	[0.5, 1.5]	[0.3, 0.7]
Period jitter	σ_ξ	0.1	0.1
Phase ratio	$[\rho^{\text{lo}}, \rho^{\text{hi}}]$	[0.55, 0.8]	[0.7, 0.9]
Offset spread	σ_Δ	0.05	0.03
Down-phase offset	μ_Δ	0.3	0.8
Up noise range	s^{up}	[0.02, 0.05]	[0.01, 0.03]
Down noise range	s^{dn}	[0.01, 0.03]	[0.005, 0.02]
Noise scale-down	κ_ν	0.15	0.15
<i>Cadence drift (dominant run only)</i>			
Reversion [iter ⁻¹]	θ		0.02
Excursion std [Hz]	σ_f	0 (swept)	
Drift floor	g_{min}		0.1
<i>Aggregate</i>			
Background counts	$N_{\text{tr}}, N_{\text{ft}}$		4, 4
Power shares	$w_{\text{main}}:w_{\text{tr}}:w_{\text{ft}}$		9 : 0.5 : 0.5
<i>Inference null</i>			
Mean serving load	B		0.5
Decode RMS	A		0.40
Decode band [Hz]	$[f_b^{\text{lo}}, f_b^{\text{hi}}]$		[0.1, 4]
MoE corr. time [s]	τ_u		5
MoE depth	m		0.4
Prefill rate [Hz]	λ_p		0.08
Prefill height	a		0.15
Prefill duration [s]	d_p		0.1
<i>Scale</i>			
Peak power [W]	P_{peak}		850
Static overhead [W]	P_0		250

Table 1. Workload-model parameters of Section 2. Training and fine-tuning values are Ko and Zhu’s (Ko & Zhu, 2025), except κ_ν , the drift parameters, and σ_ξ for fine-tuning (Ko and Zhu list it for training only; we reuse it). The inference null is ours. Ko and Zhu’s intra-phase noise is a sub-millisecond fluctuation, and each sample of our tens-of-hertz generator averages over many of them, so we scale its standard deviation by κ_ν ; the s ranges are Ko and Zhu’s values before scaling. Amplitudes are fractions of the peak power P_{peak} . The drift excursion σ_f is swept in Section 5. The scale values are ours, and are those of a single accelerator board; only the workload’s size relative to the meter’s additive terms affects detection (Section 3.1).

time series from a meter outside the operator’s control. The meter reads

$$P_{\text{meter}}[n] = (h * P_{\text{fac}})(t_n) + \eta[n], \quad (10)$$

$$t_n = n/f_s, \quad h = h_{\text{ins}} * h_{\text{del}}.$$

The meter filters the facility power, samples it at rate f_s , and adds interference and measurement noise $\eta[n]$ on its own sample grid. Round brackets denote continuous time and square brackets index the meter’s samples. The filter is a linear time-invariant system with impulse response h , which has two parts. The delivery path h_{del} is everything between the accelerators and the meter: power supplies, distribution, and any conversion or storage. The instrument h_{ins} is the meter’s own filtering: its integration window and calibration gain. Figure 2 traces the power from the workloads to the

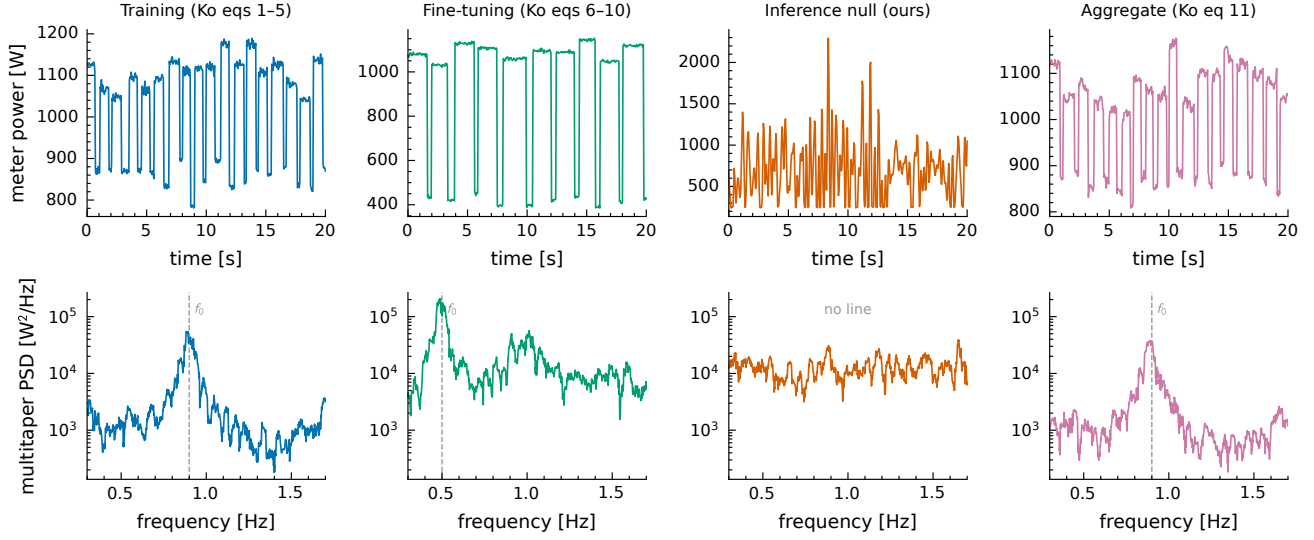


Figure 1. The four workload classes at the power meter, on the nominal channel (20 Hz sampling, no filtering or decimation, $\sigma_\eta = 4$ W). Top: a 20 s excerpt. Bottom: the multitaper spectrum over the 0.3–1.7 Hz search band, with the true cadence marked. Training, fine-tuning, and the aggregate carry a line at f_0 ; the inference null does not. Cadences are pinned for legibility ($f_0 = 0.9$ Hz training, 0.5 Hz fine-tuning); elsewhere they are drawn from the Ko bands.

verifier.

Following Ko & Zhu (2025) we assume the delivery path passes the power through unchanged, such that h_{del} is a delta function and $h_{\text{del}} * P_{\text{fac}} = P_{\text{fac}}$. The low-pass and notch of Table 2 relax this assumption in our sweeps. The notch stands for an anti-resonance of the delivery network that happens to sit near the cadence; it is the one stage that removes signal in-band while leaving the rest of the spectrum intact.

The instrument h_{ins} is the meter’s own response (Table 2). We model it as a calibration gain g followed by a trailing boxcar, so that each reading is the mean power over the preceding window τ : $h_{\text{ins}}(t) = g/\tau$ for $0 \leq t < \tau$ and zero otherwise. Its frequency response has magnitude

$$|H_{\text{ins}}(f)| = g \left| \frac{\sin(\pi f \tau)}{\pi f \tau} \right|, \quad (11)$$

which passes slow variations and attenuates fast ones, with a zero at every multiple of $1/\tau$. A window longer than about 0.6 s puts the first zero inside the 0.3–1.7 Hz search band, and a 1 s window puts it at the band centre. In the nominal channel $g = 1$ and $\tau \rightarrow 0$, so h_{ins} is also a delta function, and the meter samples at $f_s = 20$ Hz. Unlike the delivery path, the instrument is the verifier’s to choose, and Section 5.2 asks what τ and f_s must be for detection to hold.

The additive term $\eta[n]$ collects everything that reaches the readings without passing through h (Table 2). In the nominal channel it is white meter noise of $\sigma_\eta = 4$ W. Two further terms are available: a slowly wandering baseline, standing for a drifting operating point such as cooling load, and a

controller limit cycle, a square wave whose frequency itself wanders, modelled on the power-management oscillation measured on an A100 (Section 7.2). The controller is the first appearance of an instrument line: a persistent, slowly wandering spectral line that the facility’s hardware produces with no training present. Section 7.2 returns to it.

Three numbers carry units. The peak power $P_{\text{peak}} = 850$ W and overhead $P_0 = 250$ W (Table 1) are those of a single accelerator board, and the meter noise $\sigma_\eta = 4$ W is an illustrative choice, about 0.5% of the mean reading, not a measured value. None of the three is critical. P_0 is constant, so it lies at zero frequency, outside the search band. What sets detectability is the training line against everything else in the band, and at the nominal setting that is dominated by the workloads themselves: in the search band the inference null’s power spectral density exceeds that of the meter noise by about four orders of magnitude, and the training trace’s own off-line floor exceeds it by about three. Meter noise binds only when it is raised far above 4 W, which Figure 3 does over two decades.

Each trace lasts $T = 300$ s, a frequency resolution of $1/T \approx 3$ mHz. In the nominal channel every stage of Table 2 is off except the white meter noise. With the noise also off, the map returns P_{fac} unchanged and consumes no randomness. Two limits apply. The linear time-invariant form of h is a first approximation; a real delivery path may be operating-point dependent. And the model quantises only in time, through the integration window and the sample grid; it has no amplitude quantisation. Until simultaneous on-device and external-meter measurement is performed,

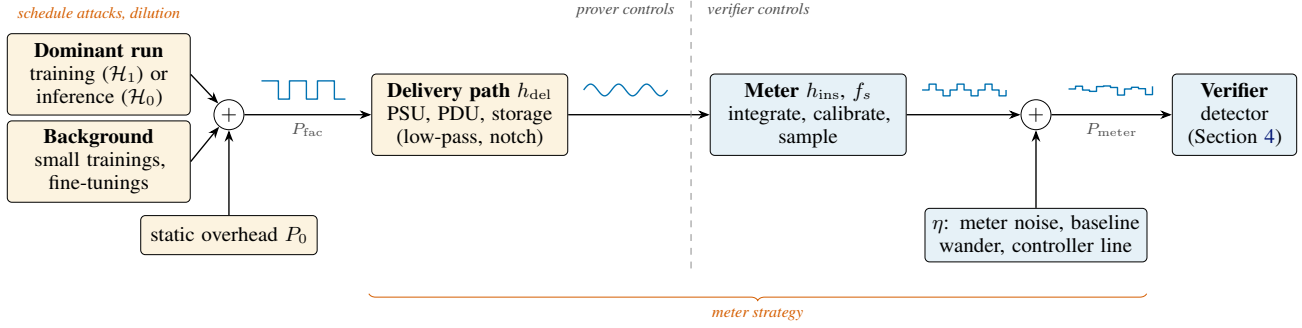


Figure 2. The measurement chain. The workloads of Section 2 and a static overhead sum to the facility power P_{fac} (Equation (9)), which passes through the delivery path and the meter before additive terms are added (Equation (10)). Sketches show one training cadence as it is smoothed, sampled, and made noisy; they are illustrative. Everything left of the dashed line is under the operator’s control (Section 3.2). Red labels mark where the adversarial strategies of Table 3 act. η lumps instrument noise with interference that physically originates behind the meter, such as the power-controller line. In the nominal channel the delivery path passes the power through unchanged ($h_{\text{del}} = \delta$) and the meter samples at 20 Hz with no integration (Table 2).

claims about what reaches an off-chip sensor are predictions (Section 7.3).

Figure 3 shows how the channel attenuates the training line. For a fixed training workload we vary one axis of Table 2 at a time and measure an effective signal-to-noise ratio at the cadence: the spectral power at f_0 over the in-band power the meter itself adds, relative to the nominal channel. Every axis lowers it, and two remove the line entirely within the range a real meter might occupy: a 1 s integration window, whose first zero falls at 1 Hz (Equation (11)), and a 1 Hz sample rate. This concerns the signal reaching the verifier, not any detector; what each detector does with what survives, and hence the minimum meter specification, is Section 5.2.

3.2. The threat model

The verifier installs and owns the power meter. The meter is tamper-evident, the verifier chooses its specification, and the operator cannot forge the meter noise η . The operator controls everything behind the meter, e.g. which jobs run, how they are scheduled, what else shares the facility, and the power-delivery path itself. Energy storage or power smoothing between the accelerators and the meter can low-pass or notch the load, and datacentre and grid operators already treat smoothing training load as an engineering problem (Ko & Zhu, 2025). We represent partial smoothing by the filtering stages of Table 2, and Section 5.2 asks how much of it the verifier can tolerate. A store large enough to flatten the load entirely would erase any passive signature; we do not model it, and discuss it among the limitations (Section 7.3). The operator knows the detector, its parameters, its threshold, and the meter specification. It does not know the realisation of η .

We consider three types of facility operator. First a *compliant* operator runs only its declared inference. A *benign* operator trains without regard to being caught by the power

meter. A *concealing* operator trains, knows the detector, and reshapes its workload to try to evade it. The concealing operator’s objective is to train without the being detected by the verifier. They use different adversarial strategies to try to subvert the verifier’s detection. Table 3 lists the adversarial strategies considered in this work. They fall into five categories: disturbing the timing of iterations, relocating the cadence, reshaping the power waveform within an iteration, mixing the run with other load, and degrading the channel. In principle the strategies may be composed, but we treat them individually. Adversarial strategies have a cost to the operator in terms of training throughput. Each strategy therefore trades concealment against cost.

Three things lie outside this threat model. First, the operator does not tamper with the meter or forge its noise. Second it does not run a semantic decoy, that is a load that reproduces the physical schedule of training without training, since manufacturing evidence of the activity it is hiding buys a concealing operator nothing. Finally, it does not move the training cadence outside the searched band or abandon synchronous iteration altogether; both are scope restrictions on the population we target (Section 7.3).

4. Detection method

In this section we present a detection method that ingests $P_{\text{meter}}[n]$, Equation (10), and decides between $\mathcal{H}_0$ and $\mathcal{H}_1$ (Section 3). The two hypotheses are separated by the iteration cadence of synchronous training. Under $\mathcal{H}_1$ the two-phase cycle of the dominant run (Equation (2)) puts a strong, narrow spectral line at its cadence f_0 . Under $\mathcal{H}_0$ the inference that replaces it spreads its power across the band with no line (Section 2.3), and the only lines left are the background’s weak ones (Section 2.4). A line at a fixed frequency has classical tests. But the cadence of a real run wanders by σ_f even without an adversary (Equation (3)),

Stage	Parameter	Nominal	Swept	Stands for
<i>Delivery path (h_{del}; the identity in the workload model)</i>				
Butterworth low-pass (order 2)	cutoff [Hz]	off	8–0.5	PSU/PDU bandwidth, smoothing
In-band notch	depth; centre [Hz]; Q	off	0.5–1; 0.6–1.2; 1.5	delivery-path anti-resonance
<i>Instrument (h_{ins} and sampling; chosen by the verifier)</i>				
Trailing boxcar integration	window τ [s]	0	0.1–2	meter averaging
Gain	g	1	—	calibration factor
Sampling (no anti-alias filter)	f_s [Hz]	20	10–0.5	meter reporting rate
<i>Additive terms (η)</i>				
Meter noise (white or AR(1))	σ_η [W]; τ_η [s]	4; white	1–120; 2	instrument noise
Baseline wander (OU)	std [W]; corr. time [s]	off	20; 60	drifting operating point
Controller limit cycle	amplitude [W pk-pk]	off	10–120 at 0.38 Hz	power-management loop

Table 2. Stages of the observation map (Equation (10)). Nominal values define the channel used everywhere unless stated. The swept ranges are those of Figure 3 and the minimum-meter sweep of Section 5.2; the hostile channels of the meter strategy (Table 3) are drawn from the same ranges. Every stage off except η is the nominal channel; every stage off including η is the identity.

and a concealing operator can make it wander further or vary the work per iteration, which defeats a fixed-frequency test cheaply (see Section 6). The detector therefore has to follow the line as it moves.

This section is organised as follows. Section 4.1 defines the detector, a Viterbi line tracker borrowed from passive sonar and continuous-gravitational-wave searches. Section 4.2 sets the threshold at which the detector declares training. Section 4.3 defines the two fixed-frequency tests we compare it against.

4.1. Viterbi line tracking

We follow the line with the hidden-Markov frequency tracking developed for passive sonar (Streit & Barrett, 1990) and carried into continuous-gravitational-wave searches (Suvorova et al., 2016; Bayley et al., 2019; Djurović & Stanković, 2011). The tracker has three parts: a spectrogram of the trace, a contrast map built from it, and a search for the best slowly varying path through that map.

Let $x[n] = P_{\text{meter}}[n] - \bar{P}_{\text{meter}}$ be the mean-removed meter trace, $N = f_s T$ samples long. We cut it into frames of M samples that overlap by half, a hop of $H = M/2$, and take the power spectrum of each:

$$S_{kj} = \left| \sum_{m=0}^{M-1} w[m] x[jH + m] e^{-2\pi i k m / M} \right|^2, \quad (12)$$

for frames $j = 0, \dots, N_t - 1$, where w is a tapered (Tukey) window and $N_t = \lfloor (N - M)/H \rfloor + 1$ is the number of frames. Bin k sits at frequency $f_k = k f_s / M$. We choose M so that each frame lasts $M/f_s = 16$ s, which gives bins $\Delta f = 1/16 = 62.5$ mHz wide, fine enough to resolve a line near 1 Hz, while leaving many frames to track over: a 300 s trace gives $N_t = 36$ frames, 8 s apart. Only the bins with $f_k \in \mathcal{B}$ enter the tracker, 23 of them in the nominal channel. Every detector in this paper searches the band $\mathcal{B} =$

$[0.3, 1.7]$ Hz. This is the training cadence band $[f^{\text{lo}}, f^{\text{hi}}] = [0.5, 1.5]$ Hz of Table 1, widened by 0.2 Hz on each side so that a cadence drawn near either edge stays in the band as it wanders.

We normalise each frame by its median over the band,

$$L_{kj} = \ln S_{kj} - \ln \tilde{S}_j, \quad (13)$$

where $\tilde{S}_j$ is the median of S_{kj} over the in-band bins of frame j . L_{kj} measures how far bin k stands above the typical in-band level of its own frame. The normalisation keeps a slow change in overall power from masquerading as a line.

The hidden state is the frequency bin occupied by the cadence in each frame. The prior that a real cadence wanders slowly enters as a penalty λ per bin of frequency change between consecutive frames. The detection statistic is the score of the best path, averaged over frames,

$$V = \frac{1}{N_t} \max_{k_0, \dots, k_{N_t-1}} \left[\sum_{j=0}^{N_t-1} L_{k_j j} - \lambda \sum_{j=1}^{N_t-1} |k_j - k_{j-1}| \right], \quad (14)$$

with $\lambda = 1$ throughout. A path pays one unit of log-contrast for each 62.5 mHz step it takes between frames, so it can follow a drift of about one bin per frame (about 8 mHz s⁻¹) if the line it follows is bright enough to pay for it. Dynamic programming evaluates the maximum over all 23^{N_t} paths exactly, in time linear in N_t . A drifting line keeps a high score because the path bends to follow it. Broadband power, such as the continuous-batching floor of the inference null, does not, because no single slowly varying path stays on top of it. Figure 4 shows both cases on the contrast map. Averaging over frames makes V comparable across traces of different length. Note that V tests whether a persistent, slowly wandering line is present, not what produced it. Accordingly, strong lines that do not have a training provenance may also trigger the detector; this is addressed in Section 7.2

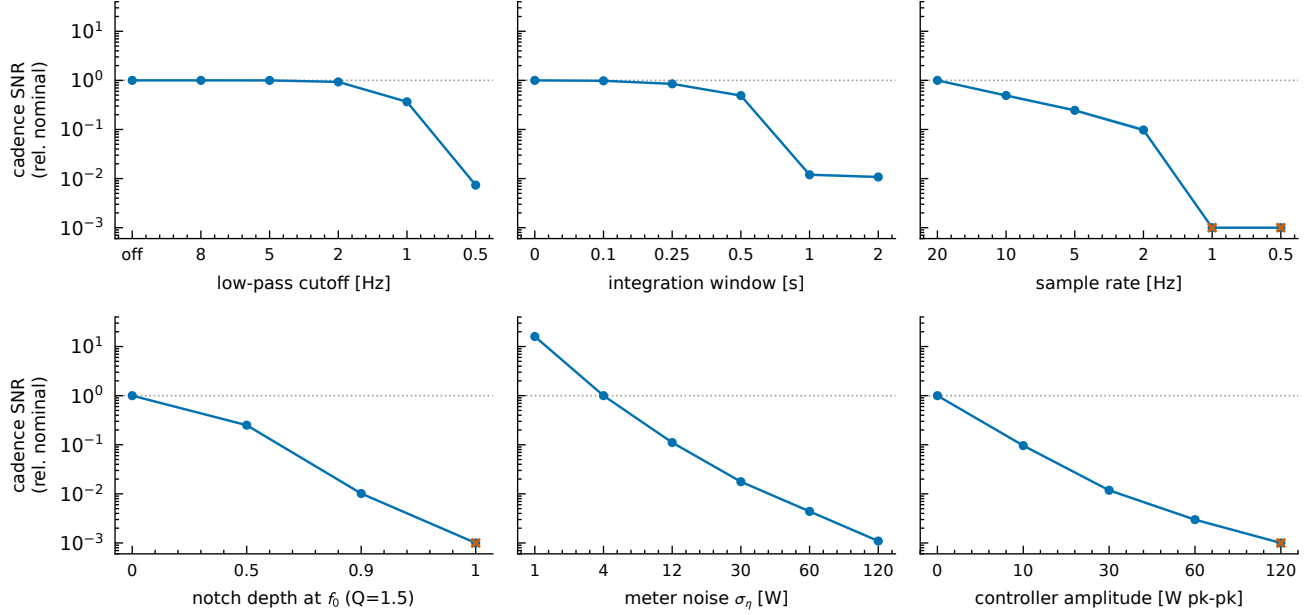


Figure 3. Sensitivity of the cadence signature to the observation map, one axis at a time on a fixed honest training workload ($f_0 = 0.9$ Hz, 20 realisations, median and interquartile range). The measure is spectral power at the known f_0 over the in-band power the meter itself contributes, normalised to the nominal channel, so that attenuating axes act on the numerator and additive ones on the denominator and every panel reads the same way: down is worse for the verifier. Crosses mark settings where the cadence is not observable at all. Note f_0 is pinned away from 1.0 Hz, which would coincide with both the 2 Hz Nyquist frequency and the first sinc null of a 1 s boxcar.

4.2. Setting the threshold

The tracker declares training when V exceeds a threshold v_α , chosen so that the false-alarm rate is α :

$$\Pr(V > v_\alpha \mid \mathcal{H}_0) = \alpha. \quad (15)$$

We use $\alpha = 10^{-2}$ unless stated. V has no analytic null distribution. We therefore set v_α empirically, as the $(1 - \alpha)$ quantile of V over simulated traces drawn under $\mathcal{H}_0$, and measure detection rate as the fraction of $\mathcal{H}_1$ traces above it. The baseline methods of Section 4.3 are thresholded the same way.

In Section 5 the $\mathcal{H}_0$ population is the null aggregate of Equation (8). Its background carries the weak lines of the small trainings and fine-tunings, so the threshold already accounts for any line that is not the dominant run’s. Calibrating on inference alone, with no background, changes the tracker’s detection rates by at most 0.02 (Appendix C).

This calibrates V against the modelled inference null. A deployed verifier has to obtain the same threshold from data it holds: from background traces known to be free of training, as continuous-wave searches set their tracking thresholds from noise-only realisations (Suvorova et al., 2016), or from a per-trace surrogate, e.g. Appendix B.

4.3. Baseline method

Two simple fixed-frequency tests serve as baselines to compare our Viterbi method against. The first is a spectral matched filter, which uses the peak of the power spectral density in the band, relative to its median over all frequencies,

$$D_{\text{MF}} = \frac{\max_{f \in \mathcal{B}} \hat{S}(f)}{\text{median}_f \hat{S}(f)}, \quad (16)$$

where $\hat{S}$ is a Welch estimate of the spectrum of $x[n]$ from half-overlapping Hann-windowed segments of 1024 samples (51.2 s at $f_s = 20$ Hz). The long segments concentrate a stationary line into one bin, which is the test’s strength; a wandering line smears across many bins and the peak collapses. The second is the Thomson multitaper harmonic F-test (Thomson, 1982; Percival & Walden, 1993), maximised over the band,

$$D_{\text{F}} = \max_{f \in \mathcal{B}} F(f), \quad (17)$$

where, at any fixed f under a stationary, locally smooth null, $F(f) \sim F(2, 2K - 2)$, with $K = 7$ tapers of time-bandwidth product $NW = 4$. $F(f)$ is the ratio of the power in a line at f to the power left around it (Equation (21), Appendix A).

Strategy	What the operator changes	Budget knob (range)	Cost anchor
<i>Timing</i>			
Period jitter	i.i.d. period jitter σ_ε of Equation (1), by inserting idle time	$\sigma_\varepsilon \in [0.05, 0.7]$	measured
Cadence drift	OU excursion σ_f of Equation (3)	$\sigma_f \in [0, 1.5]$ Hz	measured
Work variation	micro-steps per iteration G_i , drawn about $G_0 = 8$ with fractional spread σ_G ; down-phase unchanged	$\sigma_G \in [0.05, 0.7]$	measured
Phase slip	Brownian slip of iteration boundaries at fixed mean period	slip std $\in [0.01, 0.2]$ periods	—
<i>Frequency</i>			
Relocation	cadence f_0 moved toward the search-band edges	$f_0 \in [0.35, 1.65]$ Hz	—
<i>Waveform</i>			
Harmonic smoothing	raised-cosine ramp on phase transitions; comb dies, fundamental survives	ramp width $\in [0.1, 1]$ s	—
Shape filling	down-phase level raised toward the up-phase at matched mean power	fill fraction $\in [0, 1]$	measured
<i>Mixing</i>			
Dilution	training’s share of the aggregate (Equation (7)) lowered	share $\in [0.2, 0.9]$	—
<i>Channel</i>			
Meter	hostile h and η : notch at the cadence; 1 s/1 Hz integrating sampler; in-band controller; heavy coloured noise	four named channels	—

Table 3. Adversarial strategies available to the concealing operator. Each is implemented inside the generator of Section 2 and swept singly over the stated budget range in Section 6. “Measured” marks strategies with a throughput-overhead anchor from the single-A100 campaign; the others were not run on hardware and are reported without a cost (Section 6). The meter strategy degrades the channel rather than the schedule and is read as a channel requirement (Section 5.2), not priced.

5. Detection results

In this section we test the Viterbi tracker of Section 4 on synthetic meter traces. We assess if the Viterbi tracker can detect the training frequency, and what sampling settings are required by the power meter for the Viterbi tracker to work. We treat the operator as benign; an operator who reshapes the schedule to hide the line is the subject of Section 6. Under $\mathcal{H}_1$ the meter sees a dominant training run over $N_{\text{tr}} = 4$ small trainings and $N_{\text{ft}} = 4$ fine-tunings (Equation (7)). Under $\mathcal{H}_0$ it sees the null aggregate, with inference in the dominant slot over the same kind of background (Equation (8)). The shares are the defaults of Table 1, 9 : 0.5 : 0.5. Both classes therefore also carry the background’s weak lines, and the tracker has to find the dominant one among them.

This section is organised as follows. Section 5.1 measures the tracker’s detection rate as the cadence drifts, and compares it with the fixed-frequency tests of Section 4.3 and with an optimality ceiling. Section 5.2 sweeps the meter and states the minimum specification under which detection holds.

5.1. Detection rate under cadence wander

We consider a range of cadence-drift $\sigma_f \in \{0, 0.1, 0.2, 0.4, 0.8, 1.5\}$ Hz (Equation (5)) in the dominant training run, with $n = 200$ traces of 300 s per class and thresholds set at an empirical false-alarm rate of 10^{-2} . We benchmark the Viterbi method against a

Neyman–Pearson optimal ceiling (Appendix C). Figure 5 shows the result. The Viterbi tracker stays within 0.04 of the optimal ceiling across the whole σ_f axis, with detection rate 1.00 up to 0.4 Hz of drift, 0.99 at 0.8 Hz and 0.96 at 1.5 Hz. Conversely, the baselines fixed tests struggle. The spectral matched filter detects a stationary line (0.98 at zero drift) but falls to 0.59 at 0.1 Hz and 0.23 at 0.4 Hz, and the multitaper F-test has essentially no power at empirical thresholds at any drift (≤ 0.03).

5.2. A minimum meter specification

The results of Section 5.1 assume the nominal meter of Table 2: a sample rate of 20 Hz, no integration window ($\tau \rightarrow 0$), no filtering on the delivery path, and white meter noise of $\sigma_\eta = 4$ W. A real meter may sample more slowly, average over a longer window, or sit behind a delivery path that suppresses the cadence band. The meter is the verifier’s choice, so it is of interest to find the minimum specification it must meet for detection to hold. We sweep the sample rate over $f_s \in \{20, 10, 5, 2, 1, 0.5\}$ Hz and the integration window over $\tau \in \{0, 0.1, 0.25, 0.5, 1, 2\}$ s (Equation (11)). We also sweep a notch in the delivery path, i.e. a band near the cadence in which the delivery path suppresses the power swing (Section 3.1). Its centre ranges over 0.6–1.2 Hz and its depth, the fraction of the signal it removes at the centre, over 50–100%, with the meter otherwise nominal. Throughout, the honest aggregate is scored against the null aggregate, with thresholds at an empirical false-alarm rate of 10^{-2} as in Section 5.1.

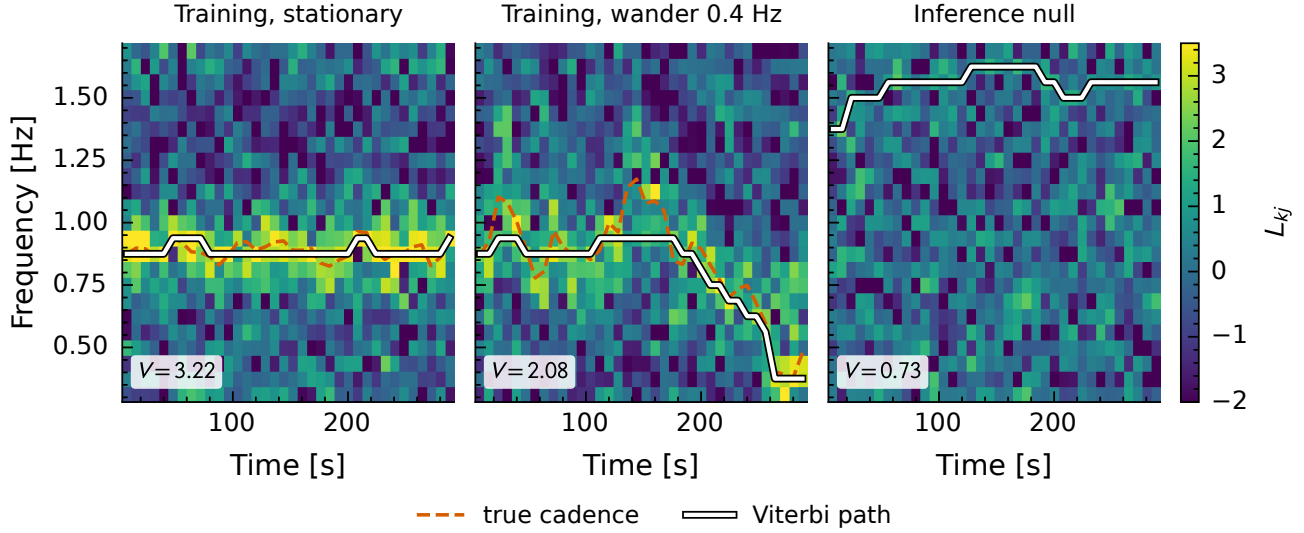


Figure 4. The Viterbi tracker on three 300 s traces at the nominal channel. Each panel is the contrast map L_{kj} of Equation (13) on the tracker’s own grid (16 s frames at half overlap, 62.5 mHz bins over $\mathcal{B}$), with the best path of Equation (14) in white and its score V inset. Dashed: the true cadence, $1/(\text{median iteration period})$ in each frame, from the generator’s iteration boundaries. Left: a stationary training cadence ($f_0 = 0.9$ Hz), which the path sits on. Centre: the same run with 0.4 Hz of cadence wander, which the path follows across most of the band; it cuts the corner off the brief excursion near 150 s, where following the line costs more in jumps than it gains in contrast. Right: the inference null, where no path finds sustained contrast and V falls to 0.73. Reproduced by `scripts/plot_viterbi_track.py`.

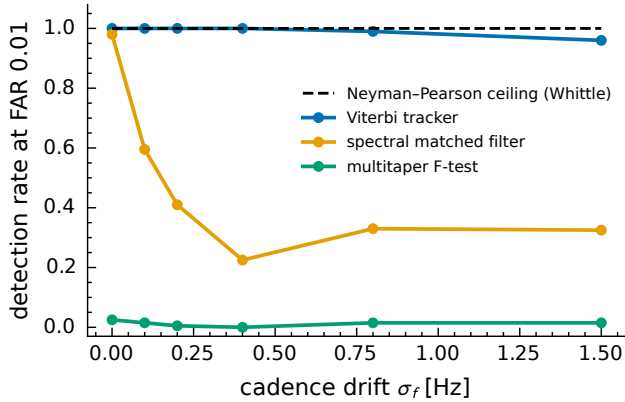


Figure 5. Detection rate at empirical false-alarm rate 10^{-2} against the null aggregate, versus cadence drift σ_f of the dominant training run ($n = 200$ per class, 300 s traces, shares 9 : 0.5 : 0.5). The Viterbi tracker stays within 0.04 of the Whittle Neyman–Pearson ceiling across the wander axis; the spectral matched filter collapses once the line moves and the multitaper F-test has no usable power. The ceiling is described in Appendix C.

The Viterbi tracker holds full power (detection rate $\gtrsim 0.9$) if and only if the meter satisfies three conditions together (Figure 6), one on each part of Equation (10). First, on the sampling, a rate of at least about 2 Hz: rates of 20, 10, 5, and 2 Hz all hold the Viterbi tracker near 1.0, 1 Hz collapses it to 0.33 as the Nyquist edge crosses the band, and 0.5 Hz is dead (every detector at chance). Second, on the instrument’s impulse response, an integration window $\tau \lesssim 0.5$ s: at

20 Hz sampling $\tau = 0.5$ s still holds (0.94), but $\tau = 1$ s puts the first zero of Equation (11) at $1/\tau = 1$ Hz, the band centre, and collapses the tracker to 0.32 even at full sample rate. Third, on the delivery path, no deep transfer-function notch at the cadence: a shallow notch, or one near the edge of the cadence band, costs little (tracker ≥ 0.80), but a notch of 90–100% depth at 1.0–1.2 Hz roughly halves tracker power (to 0.45–0.63) without erasing it (Appendix E).

Against an honest operator, then, detection needs only a meter that samples at $f_s \geq 2$ Hz, integrates over $\tau \leq 0.5$ s, and has no deep notch at the cadence.

6. Detection against a concealing operator

Section 5 assumed a benign operator, who leaves the training cadence for the tracker to find. A concealing operator can instead de-periodicise its schedule and smear the cadence away (Section 3.2). In principle this is always possible. The concealing operator can insert idle time at random between iterations, or vary the work done in each, to remove any frequency for the Viterbi detector to identify. The question is the cost in training throughput the operator must incur in order to de-periodicise enough to hide from the Viterbi detector, and in particular whether any de-periodicisation strategy is both cheap and effective.

In this section we sweep all eight schedule strategies of Table 3 and measure how much each one hides from the detectors. The operator applies each strategy to its dominant

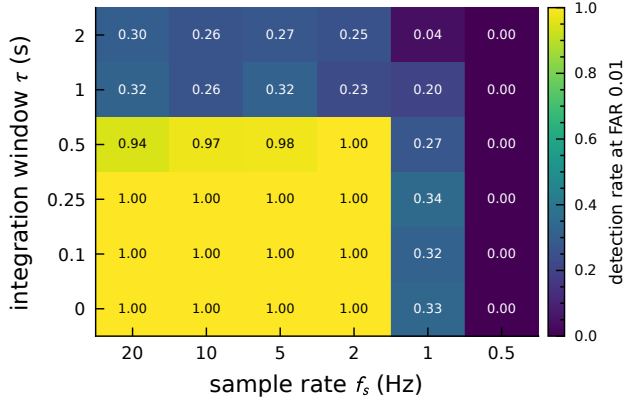


Figure 6. Minimum meter specification: Viterbi detection rate at empirical false-alarm rate 10^{-2} over sample rate $f_s \times$ integration window τ , the honest aggregate against the null aggregate ($n = 200$ per class). The live region is $f_s \geq 2$ Hz and $\tau \leq 0.5$ s.

training run, P^{main} in the aggregate of Equation (7). The background of small trainings and fine-tunings, P_j^{tr} and P_k^{ft} , stays honest, and every cell is scored against the aggregate null of Equation (8). For each strategy we step its budget knob over the range in Table 3. We measure the detection rate of the Viterbi tracker and of the fixed tests of Section 4.3, taking the better of the two fixed tests at each cell. All rates are at false-alarm rate 10^{-2} with $n = 200$ traces per class.

We measure the throughput cost incurred by four of the concealing strategies (period jitter, cadence drift, work variation and shape filling) on a single A100. We write the throughput cost as

$$c = \frac{r_{\text{honest}}}{r} - 1, \quad (18)$$

i.e. the fractional increase in wall-clock time per trained token, where r is the training throughput in tokens per second under the concealing strategy and r_{honest} that of the same run without it. For the strategies that only insert idle time, every iteration trains the same number of tokens, so c is the idle time added per iteration as a fraction of the busy time. Work variation changes the number of tokens per iteration, so there we measure r directly in tokens per second. We did not run the other four concealing strategies (phase slip, relocation, harmonic smoothing and dilution) on hardware, so they have no measured cost.

Figure 7 plots hiding (1 – detection rate) against the budget knob (and, where applicable, the measured throughput cost) of every concealing strategy Against the fixed baseline detection methods, seven of the eight strategies hide a large fraction of runs at their larger budgets; only harmonic smoothing hides nothing. Hiding mostly grows with the budget, up to 0.88 under both jitter and work variation. Only work variation hides for free. It hides 0.88 at $c \approx 0$, while jitter and drift need $c = 159\%$ and 356% to reach 0.88 and 0.71. Against the Viterbi tracker almost nothing hides. The

tracker detects at least 95% of runs in 43 of the 46 cells. It detects every run in every cell of phase slip, relocation, harmonic smoothing, shape filling and dilution, and at least 97% under drift. The three exceptions are two jitter settings and the largest work variation. Jitter at $\sigma_\xi = 0.35$ hides 0.24 for $c = 159\%$, and at 0.5 hides 0.12 for $c = 375\%$. Work variation at $\sigma_G = 0.7$ hides 0.55 at $c \approx 0$, though at every smaller σ_G it hides at most 0.05.

Work on real hardware prices de-periodicisation the same way as our measured costs, through a different channel. Four evasion strategies run on a single H200 — chunked optimizer updates, dilution with decoy inference, throttling, and diluted LoRA — cost between 28 and 69% of throughput (Gargiulo & Kulp, 2026). Each inserts or substitutes work, which is what makes it expensive, and none varies the real computation per iteration, the one strategy we find cheap.

While the Viterbi detector is robust to the majority of concealing strategies, work variation at $\sigma_G = 0.7$ is the main exception; the operator can hide 55% of 300 s observation windows at zero throughput cost. However, the strategy does have a cost in terms of training efficiency. The work variation strategy involves a few iterations accumulating many micro steps, where about one in ten accumulates 64 microsteps or more. The long blocks raise the mean micro-step count from 8 to 25, so the operator takes about a third as many optimiser steps per token as honest training, and each step on a larger batch does less per token. McCandlish et al. (2018) model an optimiser step on a batch of B samples, with the learning rate tuned for that batch, as making a fraction

$$h(B) = \frac{B}{B + B_{\text{noise}}} \quad (19)$$

of the progress of a full-gradient step, where the gradient noise scale B_{noise} sets the critical batch size. For micro-step counts G_i drawn independently, progress per sample is proportional to $\mathbb{E}[h(G)]/\mathbb{E}[G]$, so the attack needs a fraction

$$\Delta = \frac{h(G_0)/G_0}{\mathbb{E}[h(G)]/\mathbb{E}[G]} - 1 \quad (20)$$

more samples than honest training to reach the same loss, with B and B_{noise} counted in micro-batches; at $\sigma_G = 0.7$ it gives $\Delta = 53\text{--}184\%$ for B_{noise} between 8 and 128.

We also measured the training efficiency penalty for the work variation strategy directly. We trained a 30M-parameter GPT-style language model on 400M tokens of FineWeb-Edu (Penedo et al., 2024) on a single A100, three seeds per schedule. The strategy uses the same micro-batches in the same order as honest training ($G_0 = 8$), changing only how many are accumulated into each step; the gradient noise scale we measured over these runs is about 14 micro-batches. We define the learning cost as the

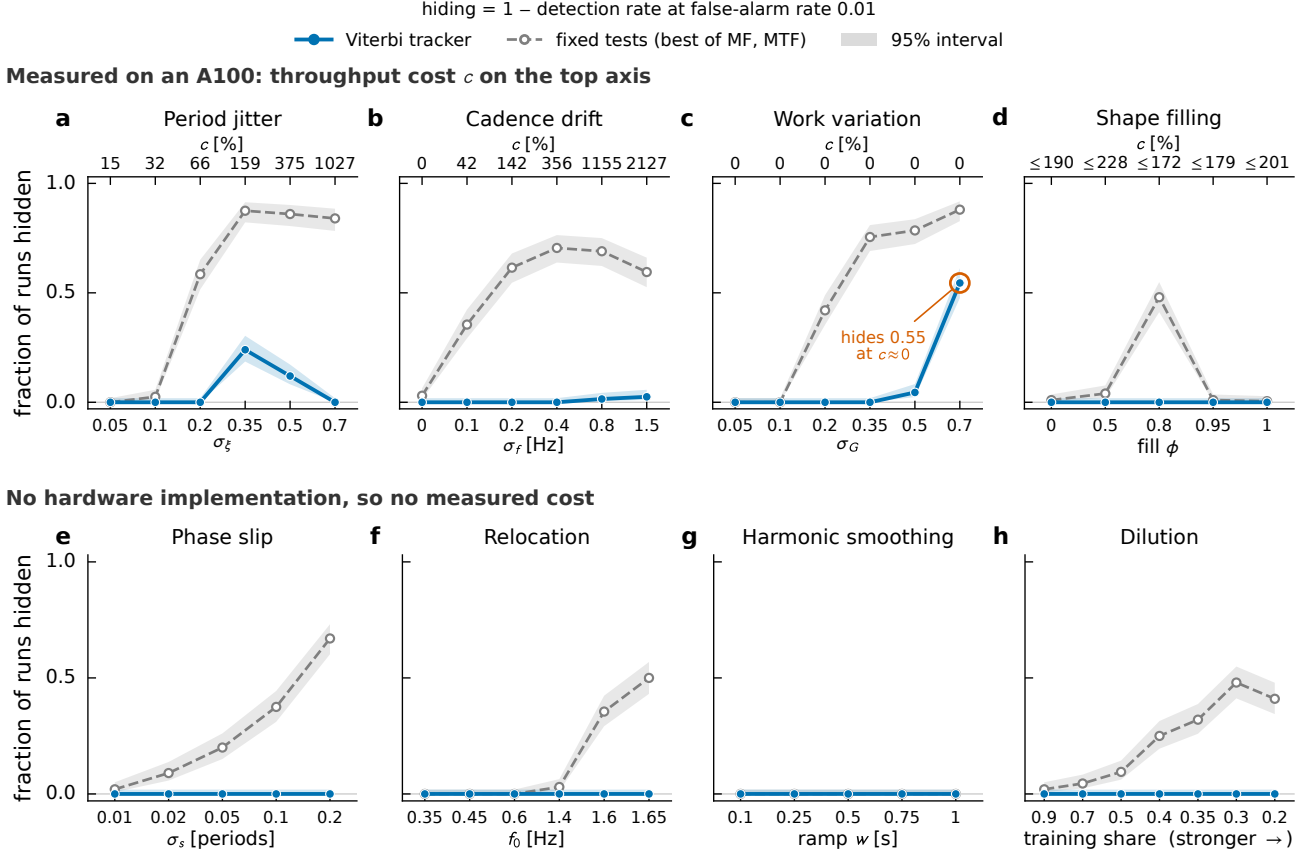


Figure 7. How much each schedule strategy hides, and what it costs. Each panel steps one strategy of Table 3 through its budget knob (bottom axis; settings evenly spaced, weakest on the left) and plots hiding, 1 – detection rate at false-alarm rate 10^{-2} , for the Viterbi tracker (blue, solid) and the fixed tests (grey, dashed; the better of the matched filter and multitaper F-test per cell). Bands are Wilson 95% intervals for $n = 200$ traces per class. **a–d**, the four strategies run on a single A100. The top axis gives the measured throughput cost c of each setting (Equation (18)); costs below 1% in magnitude, zero within noise, are shown as 0, and drift-free scheduling costs zero by definition. Shape-filling costs were measured on a jittered base and are upper bounds ($\leq$). **e–h**, the four strategies we did not run on hardware, and so with no measured cost. The tracker detects almost every cell. The exception, ringed in **c**, is work variation at $\sigma_G = 0.7$, which hides 0.55 from it at $c \approx 0$; idle-insertion jitter hides at most 0.24, for $c = 159\%$. The meter strategy degrades the channel rather than the schedule and is treated in Section 5.2.

extra tokens the strategy needs, i.e. the 400M tokens the strategy trained on, divided by the tokens honest training needs to reach the same final loss, minus one. At $\sigma_G = 0.7$ it is $267 \pm 33\%$, c.f. the predicted 163% at the measured noise scale from Equation (20). We also tried a strategy not in Table 3: draw G_i lognormally with spread s , with the mean held at the honest 8. It takes as many optimiser steps per token as honest training and never idles, so it should cost no throughput. At $s = 1.0$ it hides 0.73 from the tracker, more than the 0.55 of work variation at $\sigma_G = 0.7$, but costs $93 \pm 6\%$ in training efficiency.

7. Discussion

In this section we ask what the results of Sections 5 and 6 mean for a verifier. A Viterbi tracker detects synchronous training against a declared inference workload, given a meter

that samples at ≥ 2 Hz with ≤ 0.5 s integration. Only one kind of concealment weakens it: varying the real work per iteration, which is free in throughput but costs learning efficiency.

This section is organised as follows. Section 7.1 turns the per-window detection rates into the probability that a concealing operator is detected over a run. Section 7.2 states what a detection certifies, and how it can raise a false alarm, and Section 7.3 sets out the limitations.

7.1. From windows to runs

Every detection rate in Section 6 refers to a single 300 s window. A training run lasts days to months, and the verifier observes all of it, so what matters to a concealing operator is the probability of being detected at some point during the run. To stay hidden, the operator must evade every window,

not just a typical one.

The misses are not spread evenly across runs. Holding a run’s cadence and background fixed and drawing many windows under work variation at $\sigma_G = 0.7$, runs with $f_0 \lesssim 0.8$ Hz are detected in almost every window, while runs above about 1.1 Hz are detected in only 7–12%. Even these hardest runs are flagged at several times the null’s 1% rate. A verifier can exploit this with a run-level rule that raises an alarm when the number of detections over a run exceeds what the null would produce. At a run-level false-alarm rate of 10^{-3} , an operator whose run is detected in 7% of windows is caught with probability 0.88 after 12 hours of observation and 0.996 after a day. Concealment that works for most windows therefore does not work for a run. To stay hidden over a run, the operator must push its per-window detection rate down to the null’s, and no strategy we tested does so. [TODO: exploratory scratch check (20 runs $\times$ 20 windows, then 200 windows for the 5 least-detected runs; Viterbi only, aggregate, FAR 10^{-2}). Promote to a script with a committed results file, and repeat for the lognormal work variation of Section 6, which hides more per window. The counting rule assumes windows are independent, under the null as well: a compliant facility whose false alarms cluster would weaken it.]

7.2. What a detection certifies

A detection establishes that the trace carries a persistent, training-shaped periodic schedule. It does not establish that the computation is training. Any load that reproduces the schedule produces the same trace, and no test on the trace can separate the two (Appendix G). Such loads arise in ordinary operation: a non-ML kernel loop with a training-shaped cycle, a periodic inference request generator, or a controller limit cycle. All of them score as training under every decision rule we fitted (Appendix F). The limit cuts both ways. It caps what a positive detection proves, as well as what a mimic could fake. It is a property of the channel, not of the detector, so no better detector removes it.

Instrument lines and false alarms. The most likely such load in practice is the hardware itself. On measured single-A100 traces the spectrum contains a 0.31–0.45 Hz component, roughly ten times stronger than the training line, that appears whenever the load varies in time but not at idle or under constant load. It is the signature of the card’s own power-management loop. This is an instrument line: a persistent, slowly wandering line that the Viterbi score cannot tell from training (Section B.4). It does not help an operator hide training, but it can flag a compliant inference facility. Three things bound it. First, a known-line veto. Continuous-wave searches meet the same problem in instrumental lines, and handle it by cataloguing lines from detector characterisation data and discarding candidates coincident with them.

The analogue here is to extract several candidate paths from the contrast map (Equation (13)) and discard those that coincide with lines catalogued from the facility’s own characterisation data. We have not evaluated it, and it carries a cost a gravitational-wave veto does not: the adversary knows the veto list, and can move its cadence into a vetoed band. Second, per-card controllers need not add coherently at a facility meter. N independent controllers should grow roughly as $\sqrt{N}$, against N for a synchronous training line, and so fall relative to the training line at cluster scale. Third, a positive detection is a trigger for escalation rather than a verdict (Section 8). The order-tracked cyclostationary statistic of Appendix B partly separates the modelled controller from training by waveform shape, but only conditionally, and we do not rely on it.

7.3. Limitations

Every result in this paper is synthetic. The scenario model is literature-parameterised, not calibrated to measured hardware, so every quantitative result is a sensitivity analysis over a stated parameter family rather than a prediction. The one place we hold the model against hardware shows the gap. On a single saturated A100 the iteration cadence is present but shallow. With no communication phase, it modulates a ~ 370 W draw by only a few watts ($\sim 0.4\%$ of board power), about $75\times$ less than the scenario model assumes. This is expected. The deep compute–communication contrast that makes the cadence legible comes from synchronisation across many accelerators, so a single card is the case in which it is weakest. Testing the model where it applies needs multi-GPU training observed by an external meter alongside on-board telemetry. We plan that study, but it needs hardware and an experimental setup we do not yet have. Until then, claims about what reaches a facility meter are predictions.

The rest of the model carries narrower limitations. The observation map is a linear time-invariant first approximation. The optimality ceiling of Appendix C is optimal only under the Whittle model, hence a lower bound on the true optimum. The tracker’s thresholds are calibrated by Monte Carlo on the modelled null, not per trace, and the known-line veto of Section 7.2 is specified but not evaluated. The results are fixed-horizon. The run-level rule of Section 7.1 counts detections over a fixed number of windows and assumes they are independent, and we do not treat continuous monitoring with anytime-valid guarantees. We target synchronous, iteration-structured training whose cadence lies in the searched band; a run that moves its cadence outside the band or abandons synchronous iteration is outside the population we test. The declared workload need not be pure inference either. A facility may be permitted fine-tuning or training below a threshold, whose lines our detector would also find, and telling which run a line belongs to and how

large it is needs the integrated-energy channel we exclude.

Every robustness claim is bounded by the concealing strategies and budgets we specified; a strategy we did not model is outside it. The most important such strategy is energy storage between the accelerators and the meter. A store sized to the cadence swing can discharge during each compute phase and recharge during each communication phase, so that the meter sees a flat load. This costs the operator no throughput, so our frontier cannot price it; its costs are converters rated for the full swing, wear from cycling at the cadence, and conversion losses. Power smoothing of this kind is a grid mitigation for training load in its own right (Ko & Zhu, 2025), so it may be installed for reasons unrelated to concealment. The verifier’s remedy is placement: a meter between any storage and the accelerators sees the unsmoothed load. Finally, the one strategy that weakens the tracker is free in throughput, so its price lies in learning efficiency. We measure that price only at small scale: a 30M-parameter model, one dataset, and a limited search over concealing schedules (Section 6). Whether it holds at frontier scale, and against strategies we did not test, is the central open empirical question.

8. Conclusion

A passive power meter is a screen. Given a meter that meets the specification of Section 5.2, a Viterbi tracker detects the iteration cadence of synchronous training against a declared inference workload. Only varying the real work per iteration weakens it, and that attack, though free in throughput, still leaves a run exposed within a day and costs 93% or more in learning efficiency. The screen cannot say what the computation is: a detection certifies a training-shaped schedule, not training. A positive detection should therefore trigger escalation to an active challenge protocol (Appendix H) rather than stand as proof.

References

- Baker, M., Kulp, G., Marks, O., Brundage, M., and Heim, L. Verifying international agreements on AI: Six layers of verification for rules on large-scale AI development and deployment, July 2025. URL <https://arxiv.org/abs/2507.15916>.
- Bash, B. A., Goeckel, D., and Towsley, D. Limits of reliable communication with low probability of detection on AWGN channels. *IEEE Journal on Selected Areas in Communications*, 31(9):1921–1930, 2013. doi: 10.1109/JSAC.2013.130923.
- Bayley, J., Messenger, C., and Woan, G. Generalized application of the Viterbi algorithm to searches for continuous gravitational-wave signals. *Physical Review D*, 100:023006, 2019. doi: 10.1103/PhysRevD.100.023006.
- Bonnardot, F., El Badaoui, M., Randall, R. B., Danière, J., and Guillet, F. Use of the acceleration signal of a gearbox in order to perform angular resampling (with limited speed fluctuation). *Mechanical Systems and Signal Processing*, 19(4):766–785, 2005. doi: 10.1016/j.ymssp.2004.05.001.
- Brands, S. and Chaum, D. Distance-bounding protocols. In *Advances in Cryptology — EUROCRYPT ’93*, volume 765 of *Lecture Notes in Computer Science*, pp. 344–359. Springer, 1994. doi: 10.1007/3-540-48285-7_30.
- Cachin, C. An information-theoretic model for steganography. *Information and Computation*, 192(1):41–56, 2004. doi: 10.1016/j.ic.2004.02.003.
- Canetti, R. Universally composable security: A new paradigm for cryptographic protocols. In *Proceedings of the 42nd IEEE Symposium on Foundations of Computer Science (FOCS)*, pp. 136–145. IEEE, 2001. doi: 10.1109/SFCS.2001.959888.
- Chari, S., Rao, J. R., and Rohatgi, P. Template attacks. In *Cryptographic Hardware and Embedded Systems — CHES 2002*, volume 2523 of *Lecture Notes in Computer Science*, pp. 13–28. Springer, 2002. doi: 10.1007/3-540-36400-5_3.
- Clark, S. S., Ransford, B., Rahmati, A., Guineau, S., Sorber, J., Xu, W., and Fu, K. WattsUpDoc: Power side channels to nonintrusively discover untargeted malware on embedded medical devices. In *2013 USENIX Workshop on Health Information Technologies (HealthTech 13)*. USENIX Association, 2013.
- Dandawaté, A. V. and Giannakis, G. B. Statistical tests for presence of cyclostationarity. *IEEE Transactions on Signal Processing*, 42(9):2355–2369, 1994. doi: 10.1109/78.317857.
- Djurović, I. and Stanković, L. Viterbi algorithm for chirp-rate and instantaneous frequency estimation. *Signal Processing*, 91(5):1308–1314, 2011. doi: 10.1016/j.sigpro.2010.10.007.
- Fisher, R. A. *The Design of Experiments*. Oliver and Boyd, Edinburgh, 1935.
- Gargiulo, S. and Kulp, G. Workload identification with physical side channels for AI governance, 2026. URL <https://arxiv.org/abs/2609.00309>.
- Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., et al. The Llama 3 herd of models, 2024. URL <https://arxiv.org/abs/2407.21783>.

- Hart, G. W. Nonintrusive appliance load monitoring. *Proceedings of the IEEE*, 80(12):1870–1891, 1992. doi: 10.1109/5.192069.
- Heim, L. and Koessler, L. Training compute thresholds: Features and functions in AI regulation, 2024. URL <https://arxiv.org/abs/2405.10799>.
- Jain, R., Tran, B., Chen, K., Sinclair, M. D., and Venkataraman, S. PAL: A variability-aware policy for scheduling ML workloads in GPU clusters. In *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*, pp. 1–18. IEEE, 2024. doi: 10.1109/SC41406.2024.00032.
- Jia, H., Yaghini, M., Choquette-Choo, C. A., Dullerud, N., Thudi, A., Chandrasekaran, V., and Papernot, N. Proof-of-learning: Definitions and practice. In *2021 IEEE Symposium on Security and Privacy (SP)*, pp. 1039–1056. IEEE, 2021. doi: 10.1109/SP40001.2021.00106.
- Ko, M.-S. and Zhu, H. Wide-area power system oscillations from large-scale AI workloads, August 2025. URL <https://arxiv.org/abs/2508.16457>.
- Kocher, P., Jaffe, J., and Jun, B. Differential power analysis. In *Advances in Cryptology — CRYPTO ’99*, volume 1666 of *Lecture Notes in Computer Science*, pp. 388–397. Springer, 1999. doi: 10.1007/3-540-48405-1_25.
- Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., Gonzalez, J. E., Zhang, H., and Stoica, I. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP ’23)*, pp. 611–626. ACM, 2023. doi: 10.1145/3600006.3613165.
- Lin, J., Jiang, Z., Song, Z., Zhao, S., Yu, M., Wang, Z., Wang, C., Shi, Z., Shi, X., Jia, W., Liu, Z., Wang, S., Lin, H., Liu, X., Panda, A., and Li, J. Understanding stragglers in large model training using what-if analysis. In *19th USENIX Symposium on Operating Systems Design and Implementation (OSDI 25)*, 2025. URL <https://arxiv.org/abs/2505.05713>.
- McCandlish, S., Kaplan, J., Amodei, D., and OpenAI Dota Team. An empirical model of large-batch training, 2018. URL <https://arxiv.org/abs/1812.06162>.
- Monfared, S. K., Ganji, F., Holcomb, D., and Tajik, S. Timing and memory telemetry on GPUs for AI governance, 2026. URL <https://arxiv.org/abs/2602.09369>.
- O’Gara, A., Kulp, G., Hodgkins, W., Petrie, J., Immler, V., Aysu, A., Basu, K., Bhasin, S., Picek, S., and Srivastava, A. Hardware-enabled mechanisms for verifying responsible AI development. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. URL <https://arxiv.org/abs/2505.03742>.
- Pappu, R., Recht, B., Taylor, J., and Gershenfeld, N. Physical one-way functions. *Science*, 297(5589):2026–2030, 2002. doi: 10.1126/science.1074376.
- Patel, P., Choukse, E., Zhang, C., Goiri, Í., Warriar, B., Mahalingam, N., and Bianchini, R. POLCA: Power oversubscription in LLM cloud providers, 2023. URL <https://arxiv.org/abs/2308.12908>.
- Patel, P., Choukse, E., Zhang, C., Goiri, Í., Warriar, B., Mahalingam, N., and Bianchini, R. Characterizing power management opportunities for LLMs in the cloud. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 3 (ASPLOS ’24)*, pp. 207–222. ACM, 2024a. doi: 10.1145/3620666.3651329.
- Patel, P., Choukse, E., Zhang, C., Shah, A., Goiri, Í., Maleki, S., and Bianchini, R. Splitwise: Efficient generative LLM inference using phase splitting. In *2024 ACM/IEEE 51st Annual International Symposium on Computer Architecture (ISCA)*, pp. 118–132, 2024b. doi: 10.1109/ISCA59077.2024.00019.
- Penedo, G., Kydlíček, H., Ben Allal, L., Lozhkov, A., Mitchell, M., Raffel, C., Von Werra, L., and Wolf, T. The FineWeb datasets: Decanting the web for the finest text data at scale. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2024. URL <https://arxiv.org/abs/2406.17557>.
- Percival, D. B. and Walden, A. T. *Spectral Analysis for Physical Applications: Multitaper and Conventional Univariate Techniques*. Cambridge University Press, 1993. doi: 10.1017/CBO9780511622762.
- Rahman, R. and Tajdari, S. Detecting hidden ML training with zero-overhead telemetry, 2026. URL <https://arxiv.org/abs/2606.19262>.
- Sastry, G., Heim, L., Belfield, H., Anderljung, M., Brundage, M., Hazell, J., O’Keefe, C., Hadfield, G. K., Ngo, R., Pilz, K., Gor, G., Bluemke, E., Shoker, S., Egan, J., Trager, R. F., Avin, S., Weller, A., Bengio, Y., and Coyle, D. Computing power and the governance of artificial intelligence, 2024. URL <https://arxiv.org/abs/2402.08797>.
- Shavit, Y. What does it take to catch a Chinchilla? Verifying rules on large-scale neural network training via compute monitoring, 2023. URL <https://arxiv.org/abs/2303.11341>.

- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q. V., Hinton, G., and Dean, J. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *5th International Conference on Learning Representations (ICLR 2017)*, 2017. URL <https://arxiv.org/abs/1701.06538>.
- Streit, R. L. and Barrett, R. F. Frequency line tracking using hidden Markov models. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 38(4):586–598, 1990. doi: 10.1109/29.52700.
- Suvorova, S., Sun, L., Melatos, A., Moran, W., and Evans, R. J. Hidden Markov model tracking of continuous gravitational waves from a neutron star with wandering spin frequency. *Physical Review D*, 93:123009, 2016. doi: 10.1103/PhysRevD.93.123009.
- The White House. Executive order 14110: Safe, secure, and trustworthy development and use of artificial intelligence. Federal Register 88 FR 75191, 2023. URL <https://www.federalregister.gov/documents/2023/11/01/2023-24283/>.
- Thomson, D. J. Spectrum estimation and harmonic analysis. *Proceedings of the IEEE*, 70(9):1055–1096, 1982. doi: 10.1109/PROC.1982.12433.
- Wu, T., Wang, W., Yu, Y., Yang, S., Wu, W., Duan, Q., Yang, G., Wang, J., Qu, L., and Zhang, L. FALCON: Pinpointing and mitigating stragglers for large-scale hybrid-parallel training, 2024. URL <https://arxiv.org/abs/2410.12588>.
- Yu, G.-I., Jeong, J. S., Kim, G.-W., Kim, S., and Chun, B.-G. Orca: A distributed serving system for transformer-based generative models. In *16th USENIX Symposium on Operating Systems Design and Implementation (OSDI 22)*, pp. 521–538. USENIX Association, 2022. URL <https://www.usenix.org/conference/osdi22/presentation/yu>.
- Yücek, T. and Arslan, H. A survey of spectrum sensing algorithms for cognitive radio applications. *IEEE Communications Surveys & Tutorials*, 11(1):116–130, 2009. doi: 10.1109/SURV.2009.090109.
- Zoha, A., Gluhak, A., Imran, M. A., and Rajasegarar, S. Non-intrusive load monitoring approaches for disaggregated energy sensing: A survey. *Sensors*, 12(12):16838–16866, 2012. doi: 10.3390/s121216838.

A. Fixed-frequency baselines

This appendix gives the two fixed-frequency comparators of Section 4 in full.

When the cadence is stationary, a repeated iteration structure appears as a line, or a harmonic comb, in the power spectrum, and the natural test is the Thomson multitaper harmonic F-test (Thomson, 1982; Percival & Walden, 1993). Taper the mean-removed trace with K discrete prolate spheroidal sequences $v_k[n]$ of time–bandwidth product NW , and let $y_k(f)$ be the Fourier transform of $v_k[n] x[n]$ and $U_k = \sum_n v_k[n]$ the k -th taper’s gain at zero frequency. The least-squares amplitude of a line at f and the harmonic F statistic are

$$\hat{\mu}(f) = \frac{\sum_k U_k y_k(f)}{\sum_k U_k^2}, \quad F(f) = (K-1) \frac{|\hat{\mu}(f)|^2 \sum_k U_k^2}{\sum_k |y_k(f) - \hat{\mu}(f) U_k|^2}, \quad (21)$$

the power in the fitted line over the power the tapers see around it. We use $K = 7$ and $NW = 4$, on a zero-padded frequency grid. At a fixed frequency f the statistic $F(f)$ is distributed as $F(2, 2K - 2)$ under a locally smooth Gaussian spectral null with K tapers, so the test is *approximately pivotal under a stationary, locally smooth spectral null*. It is not an exact false-alarm guarantee across arbitrary external power traces: coloured baselines, controller interference, and co-resident load all violate the local-smoothness premise. Because the operator’s cadence is not known a priori, the test must search a band of candidate frequencies and harmonics, which requires a search correction.

The analytic correction is itself unreliable. Sweeping the padded frequency grid and applying a Bonferroni correction over the Rayleigh-resolution cell count is *anti-conservative by a factor of roughly 3.6–6×*, uniformly across our null suite: the maximum of F over the padded grid behaves like about five times more effective independent tests than the Rayleigh count, because the F ratio fluctuates below the Rayleigh scale. The analytic multitaper tail is therefore usable only as a ranking statistic; its operative false-alarm calibration is the Monte-Carlo one supplied by the harness of Section B.3. A spectral matched filter (peak PSD at the searched cadence) serves as the simple comparator. Both are honest baselines for a stationary line, and both fail the moment the line wanders: as the comparison of Section 5.1 shows, the fixed multitaper F-test retains essentially no usable detection power once the cadence drifts, which is exactly why the adaptive setting needs a different detector.

B. Order-tracked cyclostationary attribution and null validity

The main text detects training with the Viterbi score alone. This appendix reports a longer pipeline that the main text does not rely on: the Viterbi path defines a de-warp of the trace, the de-warped trace is tested for cyclostationarity at the tracked order, and the result is calibrated per trace against a surrogate null. Its value is attribution. Phase-coherent cyclic structure at the tracked order partly separates training from a power-controller limit cycle, which the Viterbi score cannot (Section B.4), but only when thresholds are set against a controller population, and the controller still inflates its surrogate-calibrated false-alarm rate (Section B.3).

B.1. Tracked cyclostationary detection

A concealing operator breaks the benign baselines cheaply by letting the cycle frequency wander, smearing the line across the searched band. We restore coherence by estimating the wander and undoing it before testing. The detector has three components, each with one job. First, the best path of the Viterbi tracker (Equation (14)), read as a function of time, estimates the instantaneous cycle frequency $\hat{f}_0(t)$; here the tracker is a nuisance-parameter estimator, *not* the detector. Second, the phase estimate

$$\hat{\theta}(t) = 2\pi \int_0^t \hat{f}_0(s) ds \quad (22)$$

defines a resampling of the trace at equal increments of $\hat{\theta}$ (tachless order tracking (Bonnardot et al., 2005)), so a wandering line becomes a constant-order line. Third, the de-warped signal is tested for cyclostationarity at candidate cycle frequency α with the Dandawaté–Giannakis statistic (Dandawaté & Giannakis, 1994)

$$Q_\alpha = N \hat{r}_\alpha^H \hat{\Sigma}_\alpha^{-1} \hat{r}_\alpha, \quad (23)$$

where $\hat{r}_\alpha$ collects the estimated cyclic autocovariances at lags τ and $\hat{\Sigma}_\alpha$ is a consistent estimate of their long-run covariance. For a fixed α , Q_α is asymptotically χ_{2L}^2 under the structural null, given weak dependence, consistent covariance estimation, and regularity.

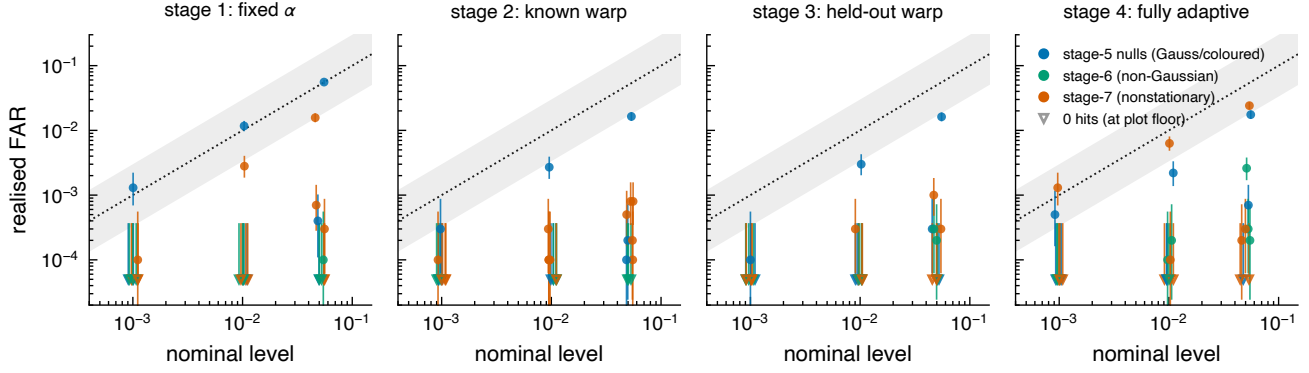


Figure 8. Realised versus nominal false-alarm rate for the adaptive pipeline (Dandawaté–Giannakis with a Bartlett HAC covariance), one panel per stage, every null in the suite, with Clopper–Pearson 95% intervals; nulls are coloured by stage group. No stationary null exceeds the nominal rate at any stage: the two-sided miss is *deflation*, never *inflation*.

Those conditions do not automatically cover the complete adaptive pipeline. The tracker performs an effective search over frequency paths (a look-elsewhere effect); the time warp is estimated from the same record it tests; the covariance estimator $\hat{\Sigma}_\alpha$ must be chosen; and the asymptotic χ^2 tail may be a poor guide at finite length. Whether the null survives these steps — and in what form it can be claimed — is the make-or-break question of Section B.3. We do not call the adaptive pipeline an analytic CFAR detector; Section B.3 establishes what it is instead.

B.2. Surrogate calibration

Because the asymptotic tail cannot be relied on for the complete pipeline — Section B.3 shows why — we calibrate each trace against a surrogate null generated from the trace itself. Given an observed record we form S surrogates by taking its discrete Fourier transform, replacing each phase with an independent uniform draw while preserving the modulus, and inverting. The resulting series has the observed record’s power spectrum and no phase structure, so the ensemble encodes the invariance “linear and stationary with this spectrum”. The *entire* pipeline — tracker, resampling, and test — is then rerun on each surrogate, and the reported p -value is the rank of the observed statistic in that ensemble. Running the full pipeline inside the surrogate loop is the point: it is what makes the tracker’s path search and the estimated warp part of the null rather than unmodelled extra freedom.

Two properties of this construction are worth stating in advance, because they determine what the resulting test can and cannot claim. It is exact only against the invariance it encodes, so a null that is genuinely nonlinear or genuinely cyclostationary is not covered — and a controller limit cycle is exactly such a null. And it is expensive: each trace costs S additional full pipeline evaluations, which is why the surrogate campaigns below are sized smaller than the analytic ones.

B.3. Null validity of the complete pipeline

The question is whether the adaptive pipeline has a null we can defend. We answer it with a staged design that isolates each adaptive ingredient in turn, crossed with a graded suite of nulls, and measured by the realised false-alarm rate at operationally small nominal levels — not only 0.05. The four stages add one adaptive step at a time: (1) a fixed cyclic frequency α with no resampling; (2) resampling under a *known* independent time warp; (3) a warp *estimated* on held-out blocks and scored only on the complement; and (4) the fully adaptive full-trace Viterbi tracker. The null suite grades from stationary Gaussian and coloured processes (white, AR(1), a resonant AR(2)), through stationary non-Gaussian processes (heavy-tailed, skewed, squared Gaussian), to locally stationary and controller-like processes (a Van der Pol-type power-management limit cycle mimicking the measured A100 controller). Each cell draws $M = 10^4$ null traces; realised rates carry Clopper–Pearson 95% intervals. Figure 8 shows the whole grid.

Level safety holds everywhere. Across the entire campaign no stationary Gaussian, coloured, or non-Gaussian null inflates the realised rate at any stage or level: the failure mode the gate was built to catch — adaptivity silently raising the false-alarm rate — does not occur. On the white null the fixed- α stage is near nominal (0.056, 0.012, and 0.0013 at nominal 0.05, 10^{-2} , and 10^{-3}); on coloured and non-Gaussian nulls the same stage is strongly conservative (realised rate ≈ 0). Tracing the three adaptive steps on the white null, where the fixed- α stage was near nominal, isolates what each one does:

- **Resampling alone (stage 1→2)** deflates the rate by about $3.4\times$ ($0.056 \rightarrow 0.016$ at nominal 0.05). Interpolating onto the angle grid correlates adjacent samples, the Bartlett estimator reads that as extra dependence and inflates $\hat{\Sigma}_\alpha$, and Q_α shrinks. Resampling *colours* the series.
- **Estimating the warp (stage 2→3)** moves the null nowhere ($0.016 \rightarrow 0.016$): the held-out design does what it promises.
- **Path selection (stage 3→4)** adds no significant inflation ($0.016 \rightarrow 0.018$; every interval pair overlaps). Even on the resonant AR(2) null, which the Viterbi tracker provably locks onto, the fully adaptive stage fires never in 10^4 traces: locking onto a broad noise resonance does not manufacture the phase-coherent cyclic structure Q_α requires.

The adaptive pipeline is therefore level-safe end to end. What fails is strict two-sided coverage: after resampling the pipeline is uniformly conservative, and the null p -values are not uniform (Kolmogorov–Smirnov uniformity is rejected at every adaptive cell).

No analytic calibration closes the deflation. We swept the open choices — record length, lag set, taper bandwidth, search band, interpolation scheme, and the covariance estimator — selecting on null-calibration quality across the suite, never on power. Only the covariance estimator moves the deflation, and it cannot be closed for the whole ladder at once (Figure 9). Widening the Bartlett bandwidth to $b = 54$ repairs the adaptive stage two-sided (AR(1) realised 0.050/0.012/0.0016 across the three levels) — confirming that the deflation is an under-bandwidth artefact of the oscillating autocovariance of the resampled series — but the *same* bandwidth breaks the fixed- α anchor, inflating white to 0.073 at nominal 0.05. No single covariance configuration in the swept family calibrates the ladder end to end, and Kolmogorov–Smirnov uniformity is rejected at every adaptive sweep point regardless. The asymptotic χ^2 tail survives only as a level-safe, power-costing conservative bound; it is not a calibration, and we do not report the adaptive pipeline as an analytic CFAR detector.

Surrogate calibration recovers the null. Running the entire pipeline — tracker, resampling, and test — inside a Fourier-phase surrogate null (which preserves the power spectrum and randomises phases, encoding the linear-stationary invariance) recovers exact calibration where the asymptotic tail was $3\text{--}6\times$ conservative. On the stationary Gaussian nulls the realised rate covers nominal across all three pipeline stages — fixed- α , held-out-warp, and fully adaptive (at nominal $0.05/10^{-2}$: white 0.051/0.009, AR(1) 0.050/0.013 fixed- α ; white 0.052/0.013, AR(1) 0.054/0.010 fully adaptive; the resonant-AR(2) and held-out-warp cells likewise) — and Kolmogorov–Smirnov uniformity is not rejected, the property the analytic p -values never achieved.¹ This is what “calibrated” means throughout the paper: an effective score with per-trace surrogate calibration, not an analytic guarantee.

The controller marks the attribution boundary. The surrogate null encodes linear stationarity, so a genuine limit cycle — which is itself cyclostationary — must and does inflate it: on the measured-controller null the surrogate-calibrated rate rises to 0.17–0.30 at nominal 0.05. This is not a defect. The detector correctly reports coherent cyclic structure in a controller trace; calling that structure *training* is the causal-attribution step that needs the nuisance model of Section 3. The controller is the periodic physical confound that Rung 1 alone cannot exclude, and its false-alarm behaviour is exactly the ceiling the claim ladder assigns to this rung.

B.4. Power against a periodic confound

The Viterbi score cannot tell a controller limit cycle from training, and the order-tracked statistic partly can. Table 4 shows this for a single training run with no background ($n = 200$ per class, 300 s traces). Positives are scored against two negative classes, each with its own empirical threshold at false-alarm rate 10^{-2} : the modelled inference null, and a structural population of equal parts AR(1), heavy-tailed AR(1), and the measured controller. With $n = 200$ negatives the

¹Frozen on the MATS compute cluster over stages $1/3/4 \times$ eight nulls at the surrogate sizing $M = 2000$ null traces $\times S = 999$ surrogates (the level-safety and bake-off numbers above are at $M = 10^4$; surrogate cells are capped lower because each carries S inner resamples). This supersedes the earlier partial $M = 1000/S = 199$ grid. The heavy-tailed but *linear* non-Gaussian nulls stay level-safe (conservative: t -innovation AR(1) 0.037/0.005, log-normal 0.035/0.006 fully adaptive); the one exception is the *nonlinear* squared-Gaussian null, which the fully adaptive stage over-rejects mildly (0.087/0.018, $\approx 1.7\times$, below the pre-registered $3\times$ threshold). Its cause is the same linear-stationarity assumption analysed next for the controller: a Fourier-phase surrogate cannot reproduce the quadratic phase structure of a squared process and so under-spreads the null. The level-safety claim is thus scoped to linear-stationary nulls.

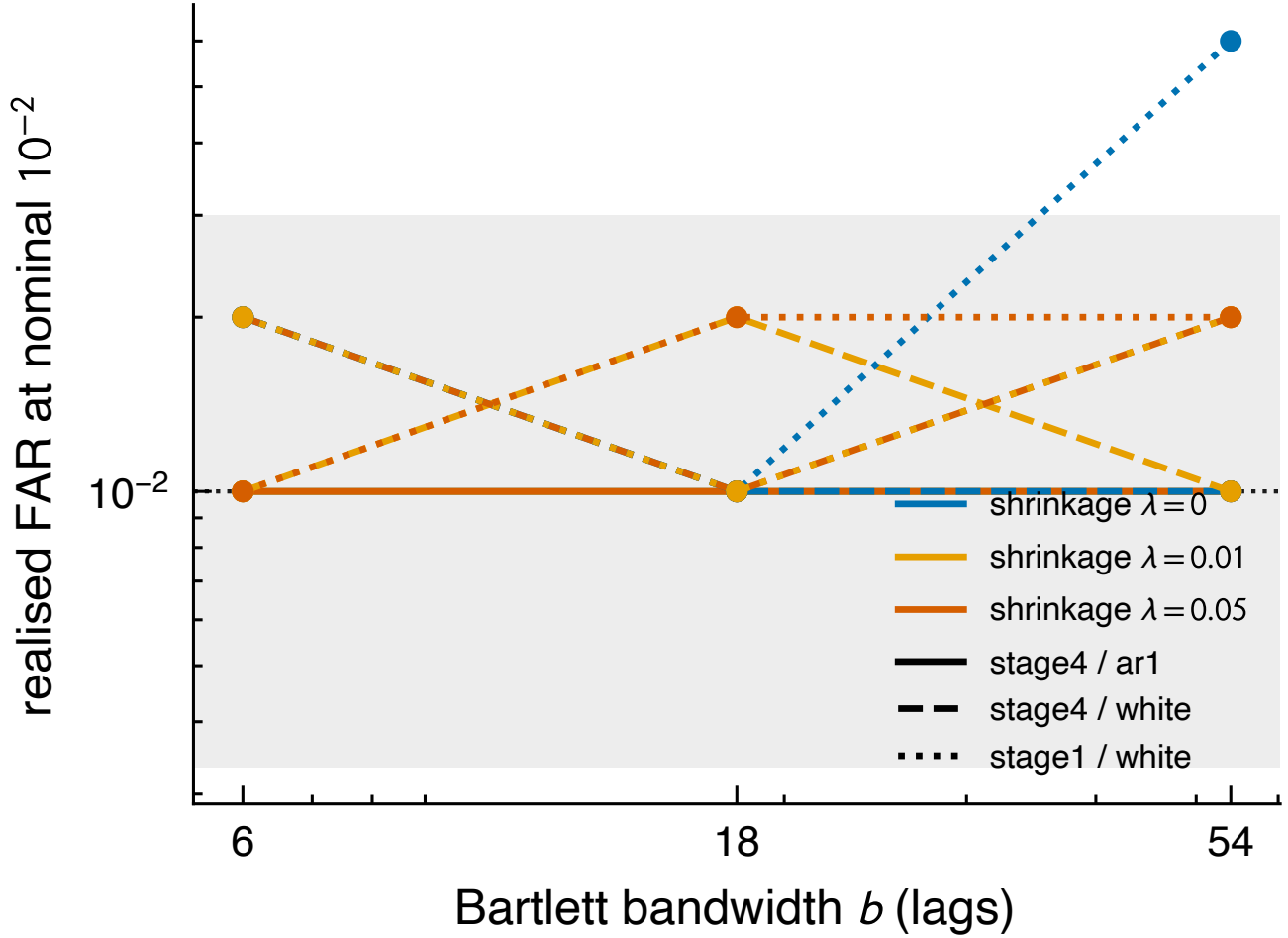


Figure 9. The covariance-estimator sweep is the fix-or-reframe axis: realised rate at nominal 10^{-2} versus Bartlett bandwidth b for each shrinkage. The bandwidth that repairs the fully adaptive stage (stage 4) inflates the fixed- α anchor (stage 1). No single configuration calibrates both.

10^{-2} threshold rests on the second-highest negative score. Against the inference null the Viterbi score is the strongest detector. Against the structural population it has no power at any drift, because a limit cycle is itself a persistent wandering line and scores as highly as training. The order-tracked statistic keeps power against the structural population at low drift, because only phase coherence at the tracked order distinguishes the two. The sharpest case is controller-only negatives at $\sigma_f = 0.4$ Hz. There the held-out-warp and fully adaptive variants detect at 0.81 and 0.91 at 10^{-2} , while the Viterbi score detects nothing.² The order-tracked power fades with drift and is gone by $\sigma_f = 0.8$ Hz. This is the separation Rung 1 can deliver against a periodic confound. It needs thresholds set against a controller population, and the scope is a single run: we did not rerun the structural negatives on the aggregate.

C. Optimality ceiling

Because we own the generators, we can benchmark the tracker against an optimality ceiling: an approximation to the Neyman–Pearson optimal test between the training and null generators. The optimal test compares how likely a trace is under $\mathcal{H}_1$ and under $\mathcal{H}_0$. Our generators can simulate traces from either class but cannot give the probability of a given trace, so we approximate. We summarise each class by its average power spectrum in the cadence band, estimated from simulated traces, and ask which spectrum better explains the periodogram of the trace (a Whittle likelihood ratio). The training cadence is unknown, so the training spectrum is averaged over a bank of 61 per- f_0 templates, each estimated from

²results/st1/bakeoff.summary.json, including the controller-only hard case.

Table 4. Detection rate at empirical false-alarm rate 10^{-2} by cadence drift σ_f (Hz), for a single training run with no background ($n = 200$ per class), against the inference null and against the structural population (AR(1), heavy-tailed AR(1), and the measured controller). The Viterbi score cannot separate training from a limit cycle; the order-tracked statistic partly can at low drift.

Detector	0	0.1	0.2	0.4	0.8	1.5
<i>vs. inference null</i>						
Viterbi	1.00	1.00	1.00	1.00	1.00	0.98
DG order, held-out warp	0.28	0.17	0.04	0.01	0.01	0.01
DG order, fully adaptive	0.73	0.62	0.55	0.23	0.06	0.01
<i>vs. structural population</i>						
Viterbi	0.00	0.00	0.00	0.00	0.00	0.00
DG order, held-out warp	0.30	0.17	0.04	0.01	0.01	0.01
DG order, fully adaptive	0.80	0.70	0.61	0.31	0.10	0.01

2000 simulated traces with drift levels drawn from the whole σ_f axis, so the ceiling knows no more about the drift than the tracker does. For the aggregate of Section 5.1 the bank fixes the dominant run’s cadence over a random background, and the null spectrum is estimated from the null aggregate. The simulated traces that build these spectra are drawn from a stream disjoint from the traces we score, and the ceiling is applied to exactly the evaluation populations of Figure 5.³

The ceiling is a benchmark, not a usable detector. It is fitted to our own generators, so it knows the exact spectra of both classes, which a verifier monitoring a real facility does not. The tracker assumes only a slowly wandering line, and the ceiling measures how much of the available power it recovers.

The ceiling detects perfectly: detection rate 1.0 at every drift (Figure 5). The training line and the line-free null are therefore separable in principle from a 300 s trace, and the tracker’s shortfall of at most 0.04, at $\sigma_f = 1.5$ Hz, is the detector’s rather than the problem’s. The natural candidate for closing it comes from the line-tracking literature. Under the hidden-Markov wander model the tracker already assumes, the Viterbi score is the best-single-path approximation, and the Neyman–Pearson statistic is the forward-algorithm marginal over *all* frequency paths (Streit & Barrett, 1990). Evaluating that statistic is future work.

The ceiling is optimal only under the Whittle model. That is a second-order, stationary-Gaussian statistic, which discards harmonic-phase coherence and non-Gaussian structure, so the true optimum can only exceed it. Since the ceiling is already perfect, this changes nothing here.

The background. The same comparison on a single training run against inference alone, with no background load, gives tracker detection rates of 1.00 up to $\sigma_f = 0.8$ Hz and 0.98 at 1.5 Hz, within 0.02 of the aggregate at every drift.⁴ At the default shares the background neither helps nor hinders the tracker.

D. Full attack sweep

The main text reports hiding against budget knob for every schedule strategy, with measured costs where they exist (Figure 7). This appendix gives the full sweep: detection against physical de-periodicisation for every strategy of Table 3 and both detector classes (Figure 10), the per-detector response to variable real work (Figure 11), and the cost-of-hiding view against each detector class (Figure 12). Here the tracking class is the best of the Viterbi tracker and the two order-tracked Dandawaté–Giannakis statistics of Appendix B; on every cell with a measured cost its envelope equals the Viterbi tracker’s alone.

The sweep runs on the aggregate of Section 5, with each strategy applied to the dominant training run only. At every knob setting it also records the physical de-periodicisation the attack induces, from the generator’s ground-truth iteration boundaries: the cadence coefficient of variation, which Figure 10 plots detection against, and the phase-diffusion coefficient. It was frozen on the MATS compute cluster (slurm array 14354, assembled by job 14355): $n = 200$ traces per class over 50 cells, seed 0, all five detectors, written to `results/st2/aggregate/frontier.summary.json`. Every rate is read at false-alarm rate 10^{-2} . The cost axis reads only the throughput overhead of the single-A100 campaign. It

³Frozen on the MATS compute cluster (slurm job 14048); `results/st1/np.ceiling.aggregate.summary.json`.

⁴`results/st1/bakeoff.summary.json` against `bakeoff.aggregate.summary.json`, same seed and $n = 200$ per class.

ST2 de-periodicisation frontier (% = measured throughput overhead)

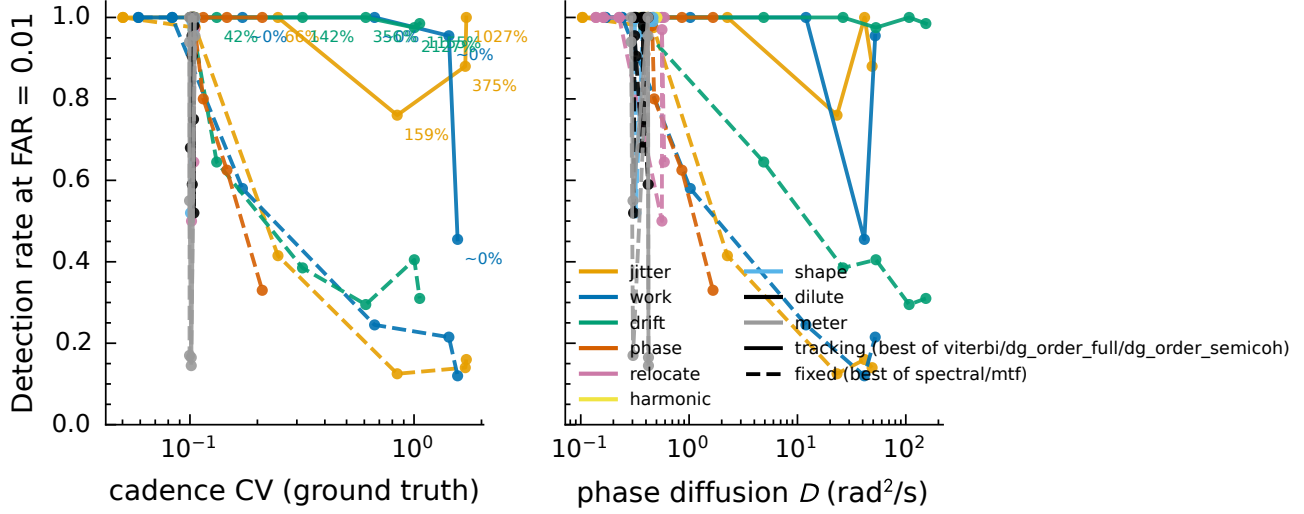


Figure 10. The de-periodicisation frontier: physical de-periodicisation (cadence coefficient of variation) against best tracking and best fixed detection rate at false-alarm rate 10^{-2} , over all adversarial strategies. The near-vertical drop of the fixed class at the variable-real-work strategy, at essentially zero throughput cost, while the tracking class holds, is the scope boundary; costlier strategies (idle jitter, drift, shaping) sit at high throughput overhead without erasing the tracked line.

does not use that campaign’s measured line amplitudes, whose transfer to a facility meter Section 7.3 questions, so the frontier does not depend on that caveat. Each overhead is the mean over eight A100 traces per level. Work variation and idle jitter at level 0.7, and drift at 0.1, 0.4, 0.8 and 1.5 Hz, were added in a later top-up on the same hardware model (data/measured_cost_anchors/README.md). The drift overheads are heavy-tailed: at 0.8 and 1.5 Hz the mean exceeds the median by a factor of 1.5 and 2. For work variation the campaign records the throughput deficit $d = 1 - r/r_{\text{honest}}$ rather than c . The two are related by $c = d/(1 - d)$ and differ by less than 0.01 percentage points at the measured $|d| < 1\%$, so we plot d directly.

The four strategies without a cost. Phase slip, relocation, harmonic smoothing and dilution were swept in the same way as the four priced strategies, but we did not run them in the hardware campaign, so they have no measured cost and appear in Figure 7 without a cost axis. They do not trouble the tracker at the levels swept: it detects every cell of all four at detection rate 1.0, dilution down to a 0.2 training share included, while the fixed tests fall as low as 0.33 under phase slip (Figure 10). A price for them would therefore only add points at zero hiding from the tracker. For relocation and harmonic smoothing a throughput price would also mislead, because their cost falls mainly on the training dynamics or on energy rather than on throughput.

E. Meter specification details

This appendix supports the minimum meter specification of Section 5.2 with two results: the notch sub-sweep, and a check that the specification does not depend on the background load.⁵

The notch. Figure 13 shows the tracker’s detection rate as a delivery-path notch moves across the cadence band of Table 1 (0.5–1.5 Hz), with the meter otherwise nominal. The notch has quality factor $Q = 1.5$ (Table 2), so it is broad: about 0.7 Hz wide at 1 Hz. A notch of 50% depth costs nothing at any centre, and one at 0.6 Hz, near the low edge of the band, costs at most 0.02. Deeper notches nearer the band centre cost more. At 0.8 Hz the tracker falls to 0.86 (90% depth) and 0.80

⁵Frozen on the MATS compute cluster (slurm array 14049), 48 cells, $n = 200$ per class, 300 s traces; results/st2/meter_boundary_aggregate_summary.json. The single-workload sweep is slurm array 5498, results/st2/meter_boundary_summary.json.

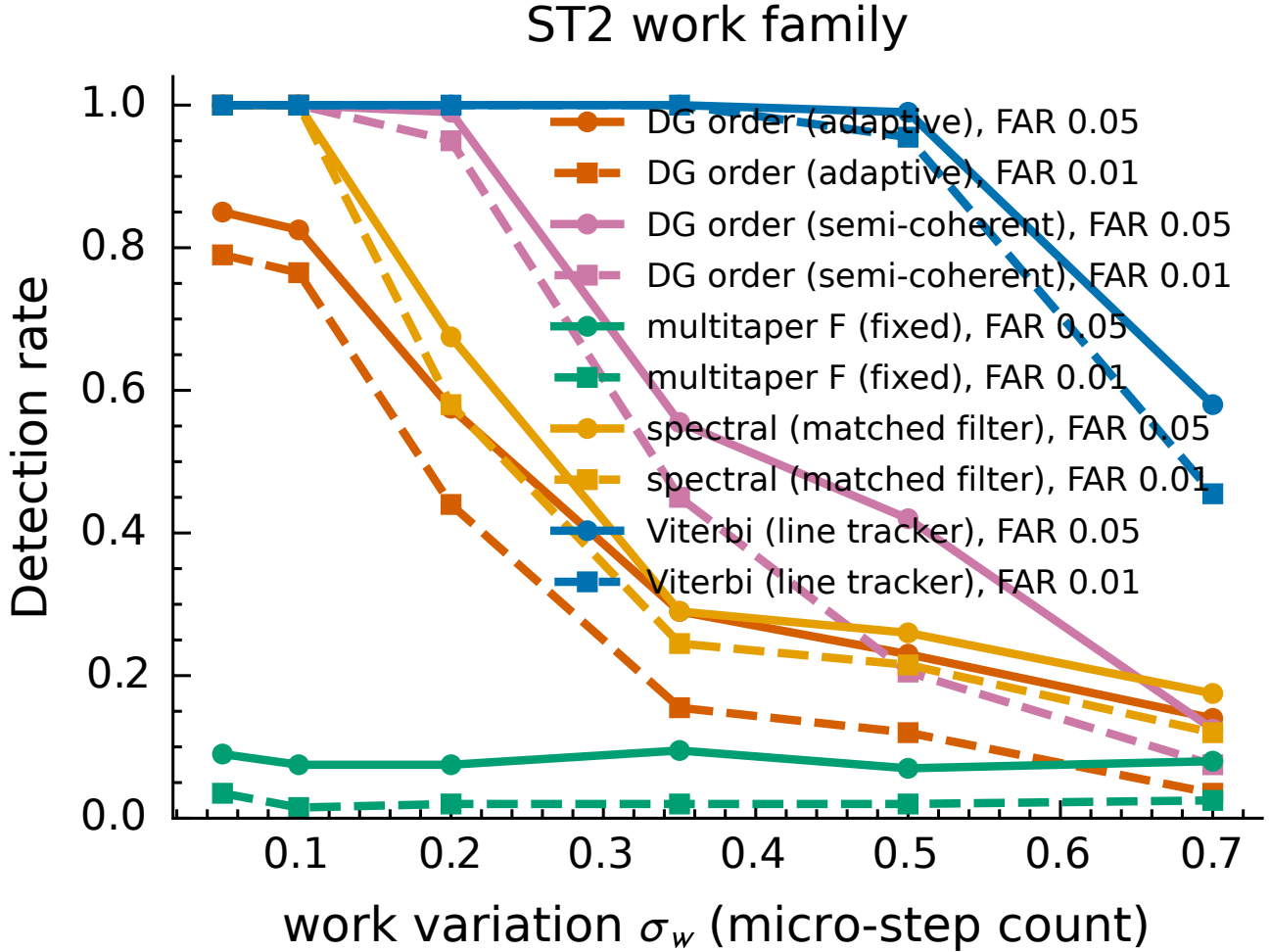


Figure 11. Detection rate versus variable-real-work level for each detector, at false-alarm rate 0.05 (solid) and 10^{-2} (dashed). By level 0.35 the fixed matched filter has fallen below chance (AUC 0.46), and the multitaper F-test stays at its floor throughout; the tracked order statistics degrade more slowly. The Viterbi tracker holds detection rate ≥ 0.95 at 10^{-2} until it bends at level 0.7 (detection rate 0.46) — a level whose measured throughput cost is zero within noise, but whose measured learning cost is 267% (Section 6).

(100%), and at 1.0–1.2 Hz to 0.45–0.63. No notch in the sweep erases detection.

The background. The same sweep on a single training run against inference alone, with no background load, changes the tracker’s detection rate by at most 0.09 in any cell and leaves the region where detection holds ($f_s \geq 2$ Hz and $\tau \leq 0.5$ s) unchanged. As for the drift sweep (Appendix C), the background at the default shares does not move the result.

F. Conditional classification against the inference null

The main text detects training against the inference null with a single statistic. This appendix reports a fuller classification task built on the same detectors — an eight-feature vector, three decision rules, six domain shifts, and seven semantic controls — whose control outcomes are summarised in Table 5.

F.1. From detection to classification

Rung 2 sharpens the structural question into a decision: is a trace more training-like than a stated inference population? A decision needs a reference, and ours is the modelled continuous-batching null of Equation (6); every Rung-2 conclusion is conditional on it. We fix the task in advance — the feature vector, the decision rules, and the reporting — so that the result is

Cost of hiding: what measured adversary cost buys against each detector class

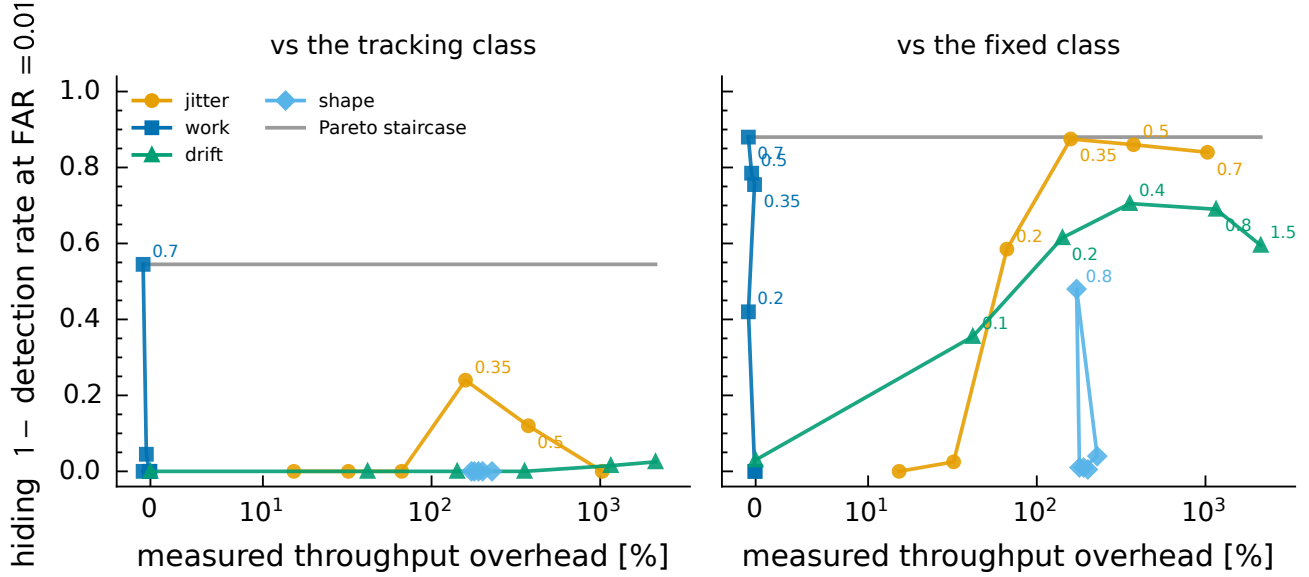


Figure 12. Cost of hiding: measured throughput overhead against hiding ($1 - \text{detection rate at false-alarm rate } 10^{-2}$) for the four priced strategies, with the Pareto staircase — the best hiding available at or below a given cost. Against both classes the staircase is flat from zero cost: variable real work at level 0.7 hides 0.55 from the tracking class (left) and 0.88 from the fixed class (right), and no attack with a throughput cost does better. Its cost is paid in learning efficiency instead, which this axis does not show (Section 6). The cost axis is linear within $\pm 10\%$ and logarithmic beyond; the four strategies without a hardware cost are not shown. Knob settings are labelled where they hide something; the rest sit on the floor.

not retrofitted to the answer.

Each trace is reduced to an eight-dimensional physics feature vector estimated *from the trace alone*, with no oracle knowledge of the cadence: the six Rung-1 detector scores (multitaper F, harmonic comb, spectral matched filter, Viterbi, fixed- α DG, and tracked-order DG), plus two stability features read off the estimated Viterbi path — its frequency-path stability and its phase-folded waveform repeatability. On this fixed vector we compare three decision rules of increasing freedom: (i) a prespecified *physics* score, a fixed linear combination with the fixed-cadence comparators pre-registered out; (ii) a logistic discriminant *fitted* to the scenario generator and scored out-of-fold; and (iii) a random-forest *learned* reference, included to expose how much separation comes from the chosen null rather than from the structural method. We report area under the ROC curve and the false-negative rate at a false-positive rate of 0.05, kept separate throughout from the Rung-1 structural false alarms.

The population the rules are evaluated on is fixed in advance too. They are fitted on the nominal setting and evaluated zero-shot on six domain shifts, each moving one factor: device headroom down to 0.70 and up to 0.95 of peak; observation duration halved to 150 s, which halves the frequency resolution; the cadence band shifted up to 0.8–1.8 Hz, so that its top now lies outside the 0.3–1.7 Hz search band; a meter carrying the controller limit cycle; and a meter with heavy coloured noise and baseline wander.

F.2. Classification against the inference null

We now apply the three decision rules of Section F.1 to the stated population, the six domain shifts, and a set of semantic controls whose training status we know.⁶

⁶Frozen on the MATS compute cluster (slurm array 5612), $n = 200$ per class, 300 s traces, band 0.3–1.7 Hz, 14 cells (the stated population, six transfer shifts, and seven semantic controls); `results/rung2/rung2-summary.json`.

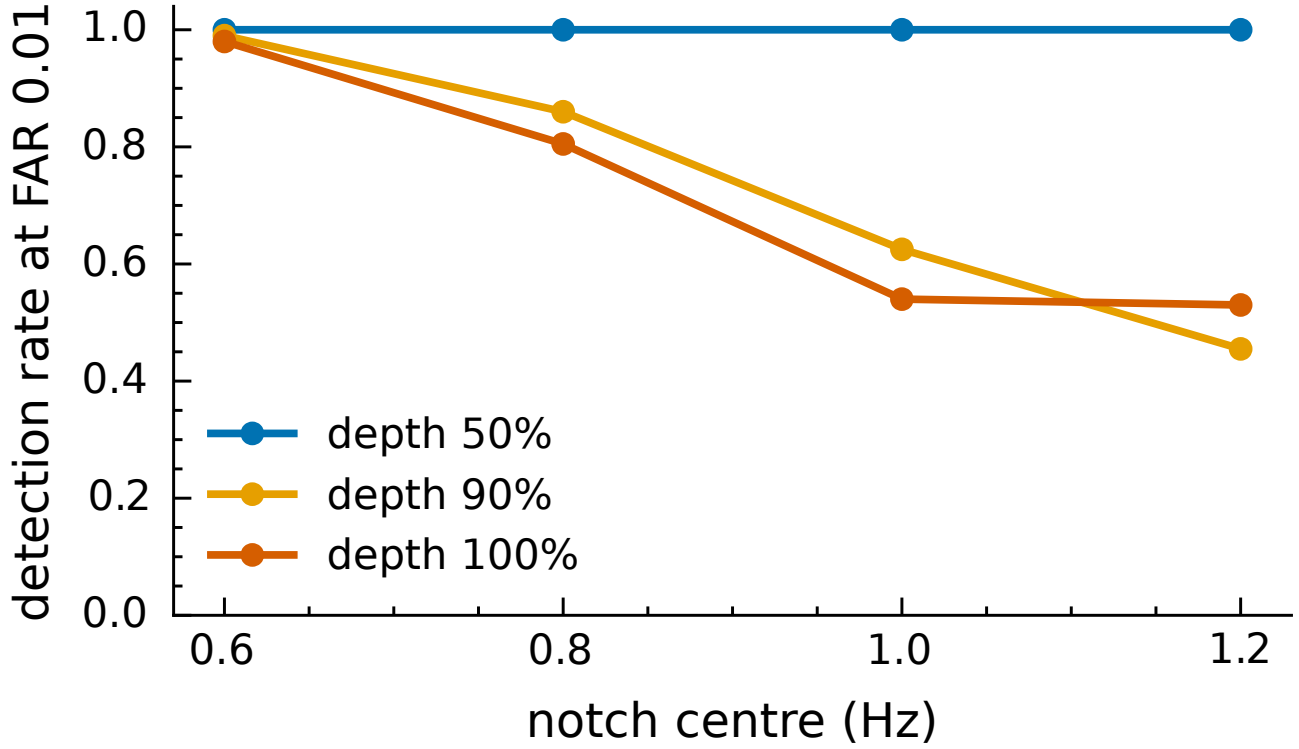


Figure 13. Notch sub-sweep: Viterbi detection rate at empirical false-alarm rate 10^{-2} versus notch centre, one line per notch depth, at 20 Hz sampling with no integration window, the honest aggregate against the null aggregate ($n = 200$ per class). A deep notch near the band centre roughly halves detection; none erases it. Reproduced by `scripts/plot_st2_meter_boundary.py --scenario aggregate`.

Stated population and transfer. On the stated population all three rules separate training from the hard inference null perfectly: area under the curve 1.00 with zero false negatives at a false-positive rate of 0.05 (Figure 14). The separation is not brittle to the population: across all six domain shifts of Section F.1, evaluated zero-shot, the area under the curve stays 1.00. The physical schedule of efficient training is far enough from the inference null that leaving the stated parameter box does not close the gap.

The semantic controls set the ceiling. The controls are the point of the rung. We built seven loads whose training status we know and asked what each rule scores (Table 5). Five non-training loads — a training-shaped non-ML kernel loop, a controller limit cycle, a periodic inference request generator, a gradient-only computation, and a discarded-update decoy that runs the full forward/backward/optimizer step and throws the update away — *all* score as training, at fraction 1.00 under every rule. A genuine training run diluted by an unrelated co-resident load is still caught (1.00). The reading is the one the claim ladder assigns to this rung: from power alone the verifier can certify a *training-shaped periodic compute schedule*, not that the computation is gradient-descent training rather than a mimic. The first three loads are the practically binding ones, because they need no adversary: the controller cycle scoring as training is the same attribution boundary Rung 1 met in Section B.3, and calling structure training needs the nuisance model of Section 3. The decoy is the extremal member of the same class and the one that cannot be engineered away — Appendix G rules out any passive detector that separates it — which is why this ceiling motivates the active protocol of Appendix H rather than a better classifier. The ceiling is a property of the observable rather than of the meter’s bandwidth. A learned classifier reading a board-level probe at 10 MHz separates training from inference by the spectral signature of the memory-bound optimizer-update phase (Gargiulo & Kulp, 2026); the discarded-update decoy runs exactly that phase and discards only its result, so the mechanism such a classifier keys on is present in a load that is not training. More bandwidth resolves the physical schedule more sharply, and it is the schedule, not the semantics, that a passive meter reads.

Table 5. Semantic-control outcomes: fraction of each load scored as training at a false-positive rate of 0.05, per decision rule. Five non-training loads all score as training; the only control the tracked rules degrade on is asynchronous (de-periodicised) genuine training, which they nonetheless mostly catch.

Control load	Training?	Physics	Fitted	Learned
Discarded-update decoy	no	1.00	1.00	1.00
Gradient-only	no	1.00	1.00	1.00
Non-ML kernel loop	no	1.00	1.00	1.00
Controller cycle	no	1.00	1.00	1.00
Periodic inference	no	1.00	1.00	1.00
Co-resident mixture	yes	1.00	1.00	1.00
Asynchronous training	yes	0.72	0.55	1.00

Asynchronous training: a robustness result, not a miss. One control, strongly de-periodicised genuine training, was pre-registered as an expected miss — the case scoped out of Rung 2. At $n = 200$ it is not missed. The tracked physics rule catches it (area under the curve 0.89, 72% flagged) and the learned reference catches all of it (100%); only the fitted linear discriminant is weak (0.75, 55%). This is consistent with the tracker-robustness story of Rungs 1 and 2: the asynchronous control de-periodicises by drifting the cadence centre and slipping the phase, which leaves a wandering but still coherent line — exactly the regime order tracking was built to survive. We therefore report it as a robustness result rather than the boundary. The genuine evasion is not slow wander but the $\approx$ zero-cost variable-real-work attack of Section 6, which moves the cadence *without* leaving a coherent line, and which is where the tracker actually bends.

G. Identifiability details

Section 7.2 states that a detection certifies a schedule, not a computation. This appendix makes that precise.

Write w for a computational execution, h for the hardware and power-delivery path, and $K_h(w)$ for the law of the observed trace that Equation (10) induces. Call two executions *observationally equivalent* when $K_{h_0}(w_0) = K_{h_1}(w_1)$. A passive verifier sees only the trace, so its decision is some measurable function of it, and on an observationally equivalent pair every such function fires with the same probability under both. No passive scalar-power detector can therefore certify the semantic content of a computation against an unrestricted adversary. A non-vacuous guarantee has to put the equivalent pairs out of scope, and there are four ways to do that within the passive channel — restricting the admissible non-training workload class, the adversary’s freedom to add or restructure computation, the hardware and power-transfer family, or the physical property claimed — alongside a fifth route that leaves the passive channel altogether by changing the protocol (Appendix H). We do not claim the list is exhaustive.

A passive verifier observes only P_{obs} , so its decision is a measurable function ϕ of that trace and its rejection probability is $\mathbb{E}_{K_h(w)}[\phi]$. If $K_{h_0}(w_0) = K_{h_1}(w_1)$ the two expectations coincide for every ϕ , so no test separates the pair, and no post-processing of the trace can (data processing cannot create distinguishability that the laws do not contain). The claim about restrictions follows by contraposition: a non-vacuous guarantee must exclude the equivalent pairs, and each listed lever excludes some of them. An amplitude degeneracy shows the pairs are not exotic. Write the power a workload draws as $r(t)F(t)$, where F is its operation rate and r the energy cost per operation. The meter constrains only the product, so a workload with a different operation rate and a compensating energy cost per operation is already observationally equivalent, and temporal structure — not amplitude — is what the trace identifies. The near-constructive same-hardware case is the discarded-update decoy: forward, backward, and optimiser kernels executed on the declared schedule with each updated state discarded reproduces the compute/communication alternation, hence the trace, without retaining a model. The converse direction is equally open to the operator: genuine training can be run asynchronously, diluted among other loads, or actively power-shaped, so the equivalence classes are populated from both sides.

H. From passive evidence to active authentication (Rungs 3–4)

The results of this paper can be read as a *claim ladder* (Table 6): four claims of increasing governance relevance, each bought at a stated cost in verifier information, and each with a hard ceiling that motivates the next. Read as a procedure, it is screening followed by escalation. Rung 1 establishes that coherent periodic power structure is present. Rung 2 converts that structure into training-likeness at the price of committing to an inference null; the detection results of this paper live here,

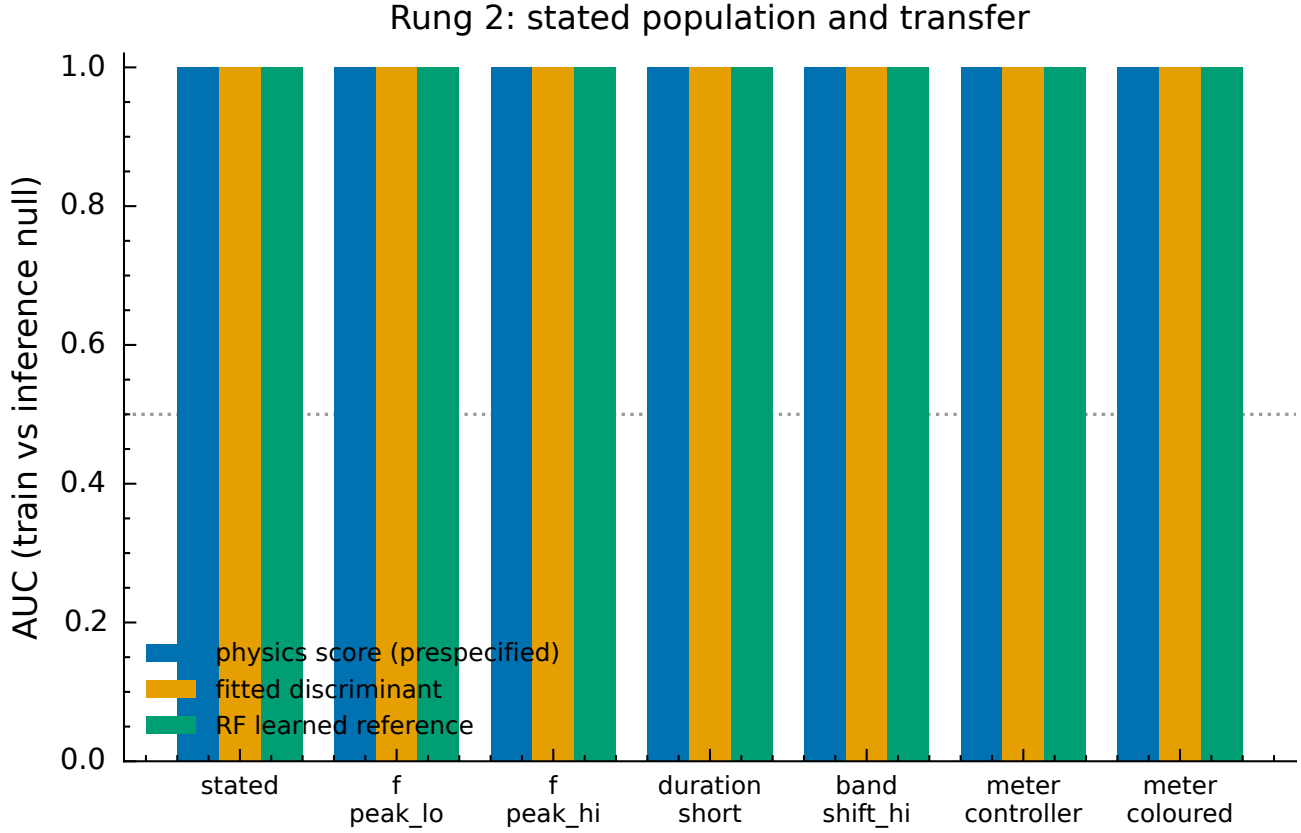


Figure 14. Rung-2 classification on the stated population (ROC / operating points for the three decision rules) and across the transfer shifts. All three rules reach area under the curve 1.00 on the stated population and hold it under every domain shift.

and its ceiling is semantic non-identifiability (Appendix G). Rungs 3 and 4 leave the passive channel for an active protocol and a transcript primitive, and are specified but not run. The ladder is a ladder of claims and assumptions, not a quantitative bound on unidentified compute.

Rung 2 certifies the physical schedule, not training semantics. Its ceiling is semantic non-identifiability: loads that reproduce the compute–communication cadence of training without training score as training under every passive rule (Section F.2), and no passive test separates them, because the meter sees only the schedule and the mimic copies the schedule. The discarded-update decoy is the sharpest instance — Appendix G rules out any passive detector that separates it — even though a concealing operator would never mount one (Section 3.2). The way past a passive ceiling is to stop watching and start asking: authenticate a physical device by its response to an unpredictable query, the premise of distance-bounding protocols (Brands & Chaum, 1994), physical unclonable functions (Pappu et al., 2002) and, in compute governance, of challenge–response telemetry (Monfared et al., 2026). If the verifier perturbs the machine and the machine must react, a mimic that merely replays a recording cannot follow: it does not know what it was asked.

Rung 3 makes this concrete with a challenge-synchronous protocol. The verifier emits an unpredictable balanced challenge sequence $c_k \in \{-1, +1\}$; the operator runs a committed, semantically-equivalent variation of its training schedule keyed to c_k — for instance a small pre-agreed shift in the boundary between two iterations that leaves the computation performed unchanged. The verifier correlates the external power trace against c_k over a causal lag window, in the manner of a lock-in amplifier or a spread-spectrum receiver recovering a known code from noise. The null needs no corpus and no matched hardware: it comes from re-randomising the challenge and recomputing the statistic. A positive result authenticates a physical fact — a load behind the meter responded to a signal it could not have predicted. This escapes the semantic decoy, whose passive schedule is by construction uncorrelated with the fresh challenge (Section 3.2). Its own ceiling is the challenge-aware dummy load: a device that receives the challenge and mimics the prescribed response without training. Unlike the passive decoy, this one is an attack an operator has reason to mount, because Rung 3 asks it to demonstrate training rather than to hide it.

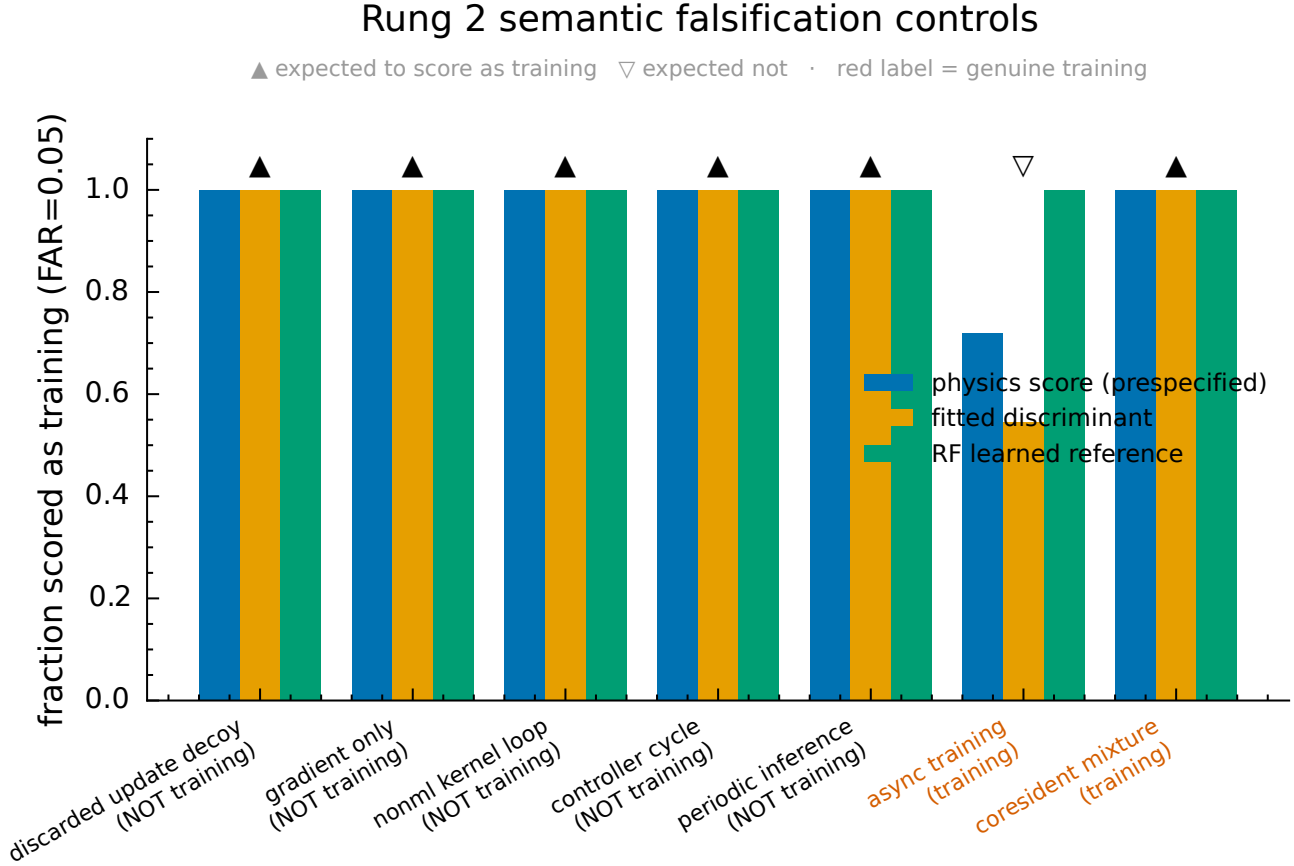


Figure 15. Semantic controls: fraction scored as training per decision rule. The non-training decoys are indistinguishable from training under every rule (the semantic-decoy ceiling); only asynchronous genuine training separates the rules, and the tracked physics rule still catches most of it.

The correlation test is exact under the sharp null that the challenge has no effect on the measured process — but only conditional on the randomisation design, and the guarantee is fragile to the usual abuses (Fisher, 1935). Any maximisation the verifier performs, over causal lags, candidate impulse responses, or preprocessing choices, must be repeated inside every re-randomised replicate, or the reported level is optimistic. Carryover between challenge periods, any adaptivity in the operator’s schedule choices, and balance constraints on the challenge sequence all belong in the design that the re-randomisation samples from. We also flag the weaker reading a careless protocol invites: over a long enough record a correlation detector accrues the $\sqrt{N}$ processing gain of any matched receiver, so a positive result must be shown to certify the challenge binding and not merely that the verifier integrated for long enough.

Rung 3 authenticates a physical response; it does not tie that response to training. Rung 4 adds that binding, and it is the rung this paper does not build. Rather than gesture at “a proof of training,” we state the missing piece as an ideal functionality (Canetti, 2001): a sound, fresh, challenge-bound transcript or proof that ties each challenge bit to the prescribed semantically-equivalent variation of declared training state transitions. Such a primitive must provide commitment before the challenge is revealed, unpredictability at commitment time, sound binding of the response to valid declared state transitions, freshness against replay, and an explicit position on whether the responding load and the declared computation are co-located. Proof-of-learning constructions (Jia et al., 2021) are the closest existing candidate, but they assume access to the training trace rather than an external meter. Conditional on such a primitive, challenge-synchronous power supplies physical evidence for the declared response; without it, the semantic work binding remains an assumed, open layer.

Two ceilings therefore remain. Rung 3’s is the challenge-aware dummy load; Rung 4’s is co-location: a genuine remote proof combined with a local challenge-correlated dummy still satisfies both the correlation test and the transcript unless the assumed primitive explicitly rules the separation out. Closing them empirically is future work, and it needs no cluster —

Table 6. The claim ladder. Each rung answers a strictly more governance-relevant question at a stated cost in verifier information, and each rung’s hard ceiling motivates the next. The fourth column attaches this paper’s evidence.

Rung	Claim	Additional requirement	re-	Evidence in this paper	Hard ceiling
1. Structural evidence	Coherent periodic power variation exists under a specified structural null	External resolved only	time-trace	Order-tracked cyclic structure is detected with surrogate-calibrated, level-safe false-alarm control on linear-stationary nulls (Appendix B); the Viterbi score alone cannot exclude a controller (Section B.4)	Periodic physical confounds; structural-null misspecification; die-to-meter attenuation
2. Conditional classification	The trace is more training-like than the stated inference population	A representative or explicitly modelled inference null		Viterbi detects training at the Neyman–Pearson ceiling across cadence wander (Section 5.1) and against moderate variable real work, which costs no throughput (Section 6), given a meter of ≥ 2 Hz and ≤ 0.5 s integration (Section 5.2); fitted rules reach area under the curve 1.00 under six domain shifts (Appendix F)	Domain shift; strong superposition, dilution or shaping; cadence moved outside the searched family; semantic non-identifiability (Table 5)
3. Active physical authentication	A load behind the meter responded to an unpredictable verifier challenge	Operator participation; a challenge-bound schedule		Specified, not run (Appendix H)	A challenge-aware dummy load; no semantic binding by power alone
4. Work verification	The physical response is tied to declared training state transitions	A sound, fresh transcript or proof and an explicit binding assumption		Assumed as an ideal functionality (Appendix H)	The remote-proof / colocation gap, unless the assumed primitive closes it

challenge modulation is legible on a single card. A full Rung-3 study would fix a concrete pair of semantically-equivalent schedules, run the complete search inside every re-randomised replicate, pit the detector against a challenge-aware dummy-load adversary, and trace detection against modulation depth, throughput overhead, and observation time. We leave it as the active-verification direction the passive results here motivate but cannot reach — the response to the ceiling that Section 7.2 makes unavoidable for any passive detector (Section 7).

I. Scratch

Material cut from the body during the 2026-09 restructure and parked here pending a decision. **This appendix is not part of the paper.**

I.1. Contributions list (cut from the introduction)

The introduction previously closed with an enumerated contributions list. It was cut because the section it headed is now prose; the claims themselves are made and evidenced in the body.

1. **The claim ladder** (Table 6) as an organising frame for passive compute verification: four claims, their information costs, and their ceilings, closed out one by one in Table 8.
2. **A tracked cyclostationary detector for Rung 1** (Section 4), with a negative methodological result we report rather than bury: no covariance configuration calibrates the complete tracker–resample–test pipeline end to end, so it is not a constant false-alarm-rate test and we do not present it as one. It is an effective score with per-trace surrogate calibration, level-exact on the linear-stationary nulls we tested. A pre-registered bake-off (Section B.4) makes the choice of detector an outcome, and against a Whittle likelihood-ratio ceiling the tracker is essentially optimal on the inference null while only the cyclostationary order family approaches the ceiling on structural confusers.
3. **The de-periodicisation frontier** (Section 6), the central result, priced against measured throughput anchors. Its shape

is an asymmetry: against fixed-frequency tests a prover reaches a hiding rate of 0.74 at essentially zero throughput cost by varying real work per iteration, whereas against the tracking class the best anchored attack reaches only 0.16 at 159% overhead.

4. **A minimum meter specification** (Section 5.2): the observation channel is a first-class condition on every rung, and we locate where each detector class dies in sample rate, integration window, and in-band transfer function. A 1 s-integrating, 1 Hz-reporting meter defeats every detector on an unmodified honest workload.
5. **Rung-2 classification with semantic falsification controls** (Section F.2). Three decision rules separate training from the stated inference null perfectly and survive six domain shifts; but five non-training loads — three of them benign operational loads, alongside a decoy that computes gradients and discards them — all score as training. The meter certifies the physical schedule, not training semantics: a ceiling no better passive detector removes, and one that limits positive evidence as much as it admits counterfeits.
6. **A negative transport case** (Section 7.3): what we measured on real hardware, why it does not carry the distributed claim, and what would falsify the prediction that it should.

Supporting these, Section 7.2 gives the formal counterpart of the Rung-1 and Rung-2 ceilings: an observational-equivalence argument for why no passive test can identify training semantics, and a covertness cost bound showing that a prover using i.i.d. phase jitter to reach a target covertness level against an exhibited verifier must pay in throughput or learning efficiency. We are explicit about what that bound does not cover, and Section 6 supplies the near-free attack that shows why the gap matters.

J. Scratch

[Parked] Old Detection-methods introduction. The weakest question on the ladder is whether the trace carries order-coherent power variation that a stated structural null does not produce. Answering it establishes *coherent repeated power structure*, not training: reading that structure as repeated multi-phase computation additionally requires the nuisance model of Section 3 that excludes periodic physical confounds, of which the measured A100 power-controller oscillation (Section 7.2) is the standing counterexample. This section develops the detectors and the calibration machinery; Section 5 reports what they do.

Two prover settings drive the design. A benign prover’s cadence is stationary, and a stationary line has a classical test. An adaptive prover lets the cadence wander, which breaks that test cheaply and forces a different construction. The benign case is easy and the paper is not about the benign case; the adaptive case is the methodological heart, and the burden here is to give its detector a defensible null.

[Parked] Old main-text detector-power subsection.

[Old heading] Detector power and how far it sits below optimal Which detector to use is an *outcome* here, not an assumption. Before scoring any signal we fixed the detector set, the conditions, and the metric, and ran a pre-registered comparison across a cadence-drift axis $\sigma_f \in \{0, \dots, 1.5\}$ Hz against two negative classes: the modelled inference null, and a mixed structural population of AR(1), heavy-tailed AR(1), and the measured controller. Section B.4 reports the full grid; three findings from it carry into everything below.

Tracking is necessary. The fixed- α Dandawaté–Giannakis statistic, handed the *true* f_0 as an oracle, collapses from detection rate 0.34 at zero drift to ≤ 0.1 under any wander — even the true frequency is useless to a coherent fixed- α test once the line moves — and the fixed multitaper F-test has essentially no usable power against either negative class at empirical thresholds. Only the tracked order family separates the wandering line from structural confounds: against the mixed structural negatives the matched filter and the raw Viterbi score have detection rate zero at every drift, because the controller third of the mixture fires them harder than the signal does. Phase-coherent cyclic structure at the tracked order is the one signature the confuser cannot fake, and this is the attribution boundary of Rung 1 drawn with power rather than with false-alarm rate. The honest limit is fast wander: beyond drift 0.8 Hz every calibratable detector fades, and a 300 s record simply does not resolve fast cadence wander against an honest null.

That ranks the detectors against each other but says nothing about how good the winner is. Because we own the generators we can also benchmark against an optimality ceiling. No closed-form likelihood exists — both classes are black-box

Table 7. The claim ladder. Each rung answers a strictly more governance-relevant question at a stated cost in verifier information; each rung’s hard ceiling motivates the next. Within each rung we consider a benign (non-adaptive) and an adaptive (hiding) prover. Table 8 closes each rung out against the evidence.

Rung	Claim	Additional requirement	Hard ceiling
1. Structural evidence	Order-coherent power variation exists under a specified structural null	External time-resolved trace only	Periodic physical confounds; structural-null misspecification; strong superposition or dilution; active power shaping; cadence moved outside the searched family; die-to-meter attenuation
2. Conditional classification	The trace is more training-like than the stated inference population	A representative or explicitly modelled inference null	Domain shift; semantic non-identifiability (training-shaped execution, benign or deliberate, without training)
3. Active physical authentication	A load behind the meter responded to an unpredictable verifier challenge	Prover participation; a challenge-bound schedule	A challenge-aware dummy load; no semantic binding by power alone
4. Work verification	The physical response is tied to declared training state transitions	A sound, fresh transcript/proof and an explicit binding assumption	The remote-proof / co-location gap, unless the assumed primitive closes it

samplers — so we use a Whittle spectral likelihood ratio on the in-band periodogram with a Monte-Carlo training template bank marginalised over cadence, fitted on a disjoint stream and applied to the exact evaluation populations. The ceiling is perfect: area under the curve and detection rate are 1.0 in every column at every drift. The training comb and the line-free nulls are separable in principle from a 300 s trace, so all the difficulty the comparison exposes lives in the detector, not in the problem.

Read as a fraction of that ceiling, the picture sharpens in a way the frontier of Section 6 depends on. Against the inference null the Viterbi tracker is essentially Neyman–Pearson optimal, holding the ceiling’s power across the whole wander axis, while the fixed matched filter’s fraction of attainable power collapses from 1.0 at zero drift through 0.78 and 0.38 to below zero by drift 0.4 Hz. Against the structural confusers the tracker is *not* enough: only the tracked order family approaches the ceiling, reaching 0.93 of it at low drift and fading to 0.10 at the fastest wander. The widening gap at high drift is honest headroom above today’s detectors rather than a limit of the channel, and the one caveat runs in our favour — the Whittle ceiling discards harmonic-phase coherence and non-Gaussian structure, so the true optimum can only exceed it.

[Parked] Old Discussion opening, ladder lead-in, ladder tables, and governance readings. We set out to ask what a passive power meter can certify about AI training, and organised the answer as a claim ladder in which each rung answers a more governance-relevant question at a higher cost in verifier information, and each rung’s ceiling motivates the next. Table 8 closes the ladder out with the paper’s evidence attached to each rung.

This section fixes what the verifier sees, who is trying to hide from it, and what can be argued about the channel before any detector is run. It also fixes the vocabulary for the claims a meter can support. We organise those claims as a *claim ladder* (Table 6): four claims of increasing governance relevance, each bought at a stated cost in verifier information, and each with a hard ceiling that motivates the next. Rung 1 establishes that order-coherent power structure is present, which is evidence rather than identification. Rung 2 converts that structure into training-likeness at the price of committing to an inference null; its ceiling is semantic non-identifiability. Rungs 3 and 4 leave the passive channel for an active protocol and a transcript primitive, and are specified but not run (Appendix H). The ladder is a ladder of claims and assumptions, not a quantitative bound on unidentified compute. Within every rung we consider two provers: a **benign** (non-adaptive) prover who runs its workload without regard to the meter, and an **adaptive** (hiding) prover who knows the detector and reshapes its schedule to evade it. Most of what is interesting lies in the gap between them.

Governance reading. A passive meter certifies efficient iteration-structured training against cheap, fixed tests only within a bounded de-periodicisation budget; a prover free to vary real work escapes those tests for nothing. Defeating the order tracker on top of that requires either a systems and learning cost we could not measure or a degradation of the meter channel itself (Section 5.2). The results of this section are therefore necessary conditions and an upper bound on hideability rather than a proof of training: they bound what a prover can hide from a fixed detector cheaply. The complementary direction — a

Table 8. The claim ladder closed out: what this paper establishes at each rung, where, and the hard ceiling it hits. Mirrors Table 7.

Rung	What the paper establishes	Evidence	Demonstrated ceiling
1. Structural evidence	Order-tracked cyclic structure is detected against stationary and controller-like nulls; the tracker is Neyman–Pearson optimal against the inference null	Section 5, Section B.4	Fast cadence wander (>0.8 Hz on 300 s); attributing structure to training needs the nuisance model; the meter must meet Section 5.2
2. Conditional classification	The trace is more training-like than the stated inference null (area under the curve 1.00, stable across domain shift)	Section F.2	Semantic non-identifiability: non-training loads with a training-shaped schedule score as training, benign ones included; Section 7.2 shows no passive detector can do better
Frontier	Fixed tests collapse under $\approx$ zero-cost variable real work while the tracker holds up to level 0.5; costly de-periodicisations buy no concealment	Section 6	At level 0.7 the tracker detects about half of runs at no throughput cost; the learning cost is unmeasured
Cost of hiding	Hiding to detectability ε requires a distortion $\sigma^*(\varepsilon)$ that the measured anchors price at tens of percent of throughput, rising above 100% (i.i.d. period jitter)	Section 7.2, Appendix G	Read off a fixed-bin verifier, so valid against any class containing it, but loose; one adversarial strategy; internal to the stated generator; the throughput-or-learning disjunction is an assumption, not a result
3–4. Active authentication	A challenge-synchronous protocol and an assumed binding primitive, specified but not run	Appendix H	Challenge-aware dummy load; the co-location gap

lower bound on what hiding must cost, which no detection sweep can supply — is available for one adversarial strategy against a fixed verifier (Section 7.2) under the cost-of-hiding assumption, and the distortion it requires can be priced against the same anchors used here. The tracking class dominates the fixed class at every cell we measured, so that threshold is a valid — but loose — reading of what a tracker would require: it is read off the weaker detector, so it understates the true requirement. Establishing it against the tracking class directly, rather than inheriting it from the measured dominance, remains open.

Governance reading. What a regulator can buy today is graded. Rung 1 buys physical evidence of an efficient iteration-structured compute schedule from an external meter alone, robust to a wander-and-drift adversary but not to a prover who varies real work or degrades the meter. Rung 2 buys the further claim that the schedule is more training-like than a stated inference baseline — but not that the computation is training rather than a schedule-matched mimic, benign or deliberate, which is a limit of the passive channel, not of the method. That limit runs both ways: it caps what a positive detection proves as well as what a counterfeit could fake. Separating training from a mimic, and binding a physical response to declared work, both require moving off the passive channel to the active protocol of Appendix H and the cryptographic primitive it assumes. The passive meter is thus a cheap, deployable front line that narrows the question and prices the cheap evasions, not a stand-alone proof of training.

The problem of

The problem of X is well established

Inferring which activities are running from one aggregate meter is the founding problem of non-intrusive load monitoring (Hart, 1992; Zoha et al., 2012), with a power-side-channel malware detector as its closest security-flavoured descendant (Clark et al., 2013). Deciding whether a source is active against a transmitter trying to stay below the detection floor is spectrum sensing, whose taxonomy of energy, matched-filter and cyclostationary detectors structures our own comparison (Yücek & Arslan, 2009).

Tracking a narrowbandline whose frequency wanders is passive sonar’s lofargram problem (Streit & Barrett, 1990), carried into continuous-gravitational-wave searches (Suvorova et al., 2016; Bayley et al., 2019), and undoing that wander before testing is tachless order tracking (Bonnardot et al., 2005).

Each of these literatures assumes something the governance setting removes:

load monitoring assumes indifferent appliances with clean switching edges;

side-channel analysis, from differential power analysis onward (Kocher et al., 1999; Chari et al., 2002), assumes a proximate high-bandwidth probe and a cooperative target;

the line trackers assume a stationary noise floor with no adversary behind it.

None of them supplies a calibrated test for a quasi-periodic, frequency-wandering signature observed through a low-rate filtered meter, and none prices what it costs the source to remove it.

The adversarial half does have an owner. Pricing concealment against detectability is the warden-versus-covert-transmitter game of covert communication (Bash et al., 2013) and the relative-entropy security measure of information-theoretic steganography (Cachin, 2004): both bound how detectable a hidden activity remains as a function of how constrained it is, and both do so for *every* verifier at once rather than for one detector. Their results assume a transmitter free to shape its signal. A training schedule is not free: its communication phase has a physical duration that does not shrink when the compute phase does, and each distortion of the cadence costs throughput or learning efficiency. That constraint is what makes a cost-of-hiding bound possible here, and no existing bound covers it.

What is missing is a method that turns a low-rate external power trace into a calibrated test for iteration-structured training, prices its evasion against an adversary who knows the test, and states what the channel cannot certify at any price. Without one, “power monitoring” enters policy proposals as a binary capability: no one can write the meter specification a regulator would have to procure, budget the evasions a prover would actually attempt, or say which governance claims a positive detection does and does not support.

and U.S. Executive Order 14110 briefly operationalised one, requiring reports of runs above 10^{26} FLOP (The White House, 2023).

Every such scheme presupposes that a regulator can tell when training is happening on monitored hardware. Three channels could tell it. Prover-owned telemetry is informative but conditional on tamper-resistant read paths that do not yet exist (Rahman & Tajdari, 2026; O’Gara et al., 2025); challenge–response protocols remove that trust anchor but require the prover’s participation (Monfared et al., 2026); and a verifier-owned probe on an accelerator’s supply conductors requires neither, at the price of physical access to the board and continuous on-site presence (Gargiulo & Kulp, 2026).

Multi GPU synchronous AI training of the type that typifies frontier AI models alternates a compute phase, a forward backward pass with high power requirements, with a communication gradient all reduce phase where the GPUs all sync and power requirements drop. [what about async training? maybe we need to mention this somewhere? perhaps in limitations?]

In practice the problem is harder as the probe is at a higher level i.e. more remote a meter three to four orders of magnitude slower than the probe, behind a power-delivery path that low-passes and dilutes the signal, reading a prover who knows it is being read.